



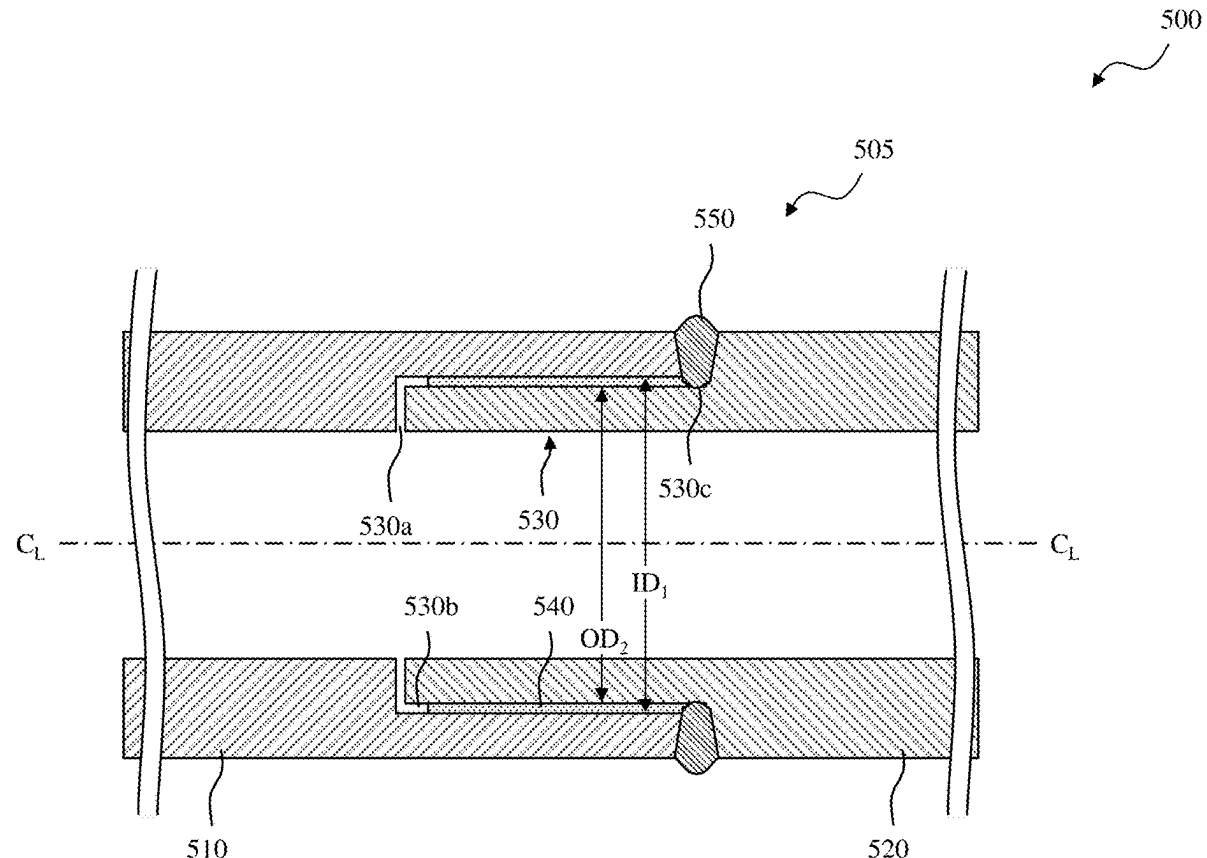
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(19) **United States**(12) **Patent Application Publication**
Saphira et al.(10) **Pub. No.: US 2025/0277548 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **DOWNHOLE TOOL WITH CRACK
COMPLIANT SEAL AND MECHANICAL
STRENGTHENING FEATURE AT A JOINT
THEREOF**(71) Applicant: **Halliburton Energy Services, Inc.**,
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Carrollton, TX (US); **David Joe Steele**,
Carrollton, TX (US)(21) Appl. No.: **19/210,231**(22) Filed: **May 16, 2025****Related U.S. Application Data**(63) Continuation of application No. 18/348,205, filed on
Jul. 6, 2023, now Pat. No. 12,326,203.(60) Provisional application No. 63/511,392, filed on Jun.
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(2013.01)

(57)

ABSTRACT

Provided is a downhole tool, a well system and a method. The downhole tool, in one aspect, includes a first member, the first member having a first member hardness value (HV_1), and a second member, the second member having a second member hardness value (HV_2), the first and second members positioned proximate one another forming an overlapping space therebetween. The downhole tool according to this aspect, further includes a mechanical strengthening feature located between the first member and the second member, the mechanical strengthening feature configured to increase an engineering rating of the overlapping space, and a crack compliant seal positioned at the overlapping space, the crack compliant seal having a compliant seal hardness value (HV_{cs}) less than or equal to one or more of the (HV_1) or the (HV_2).



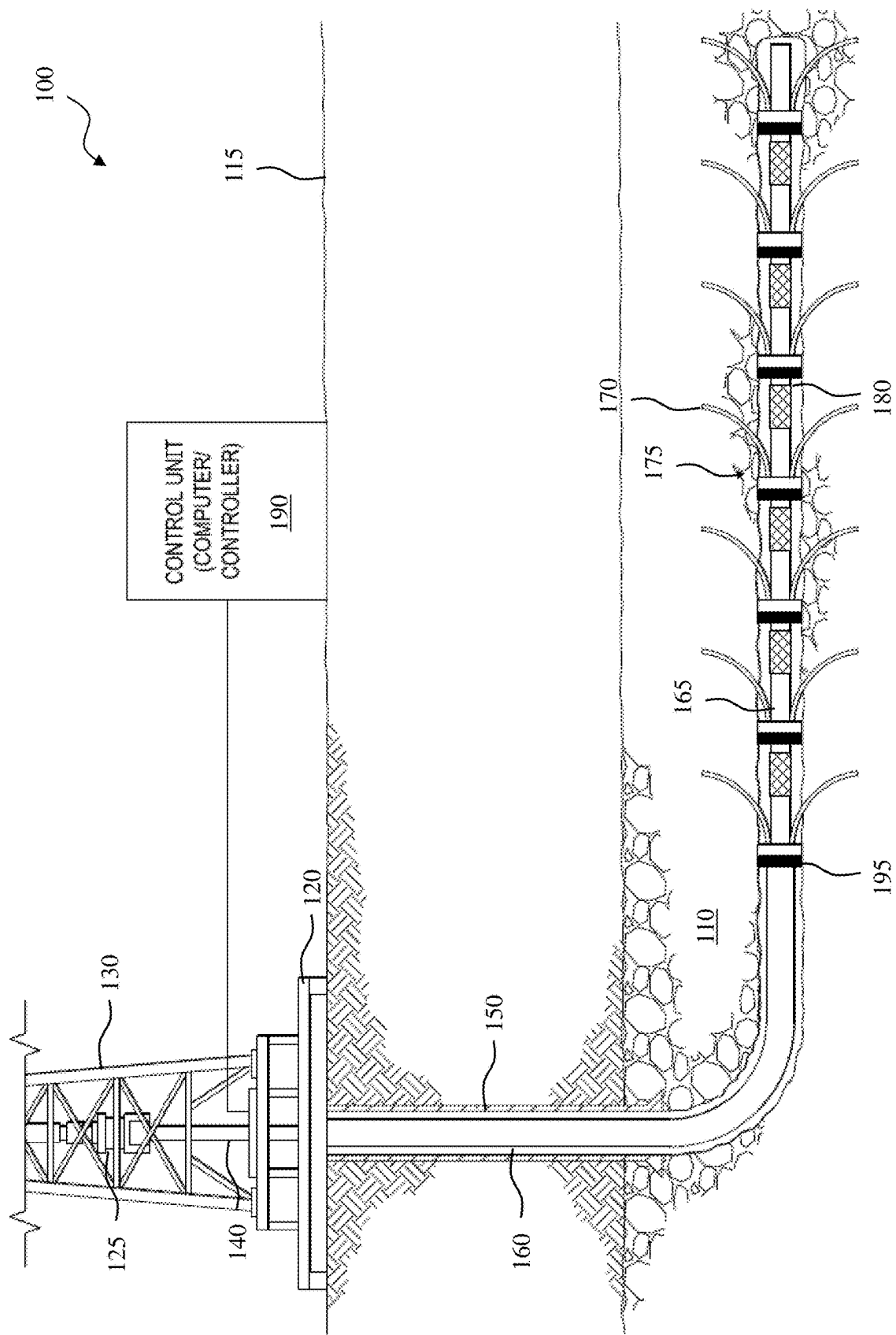


FIG. 1

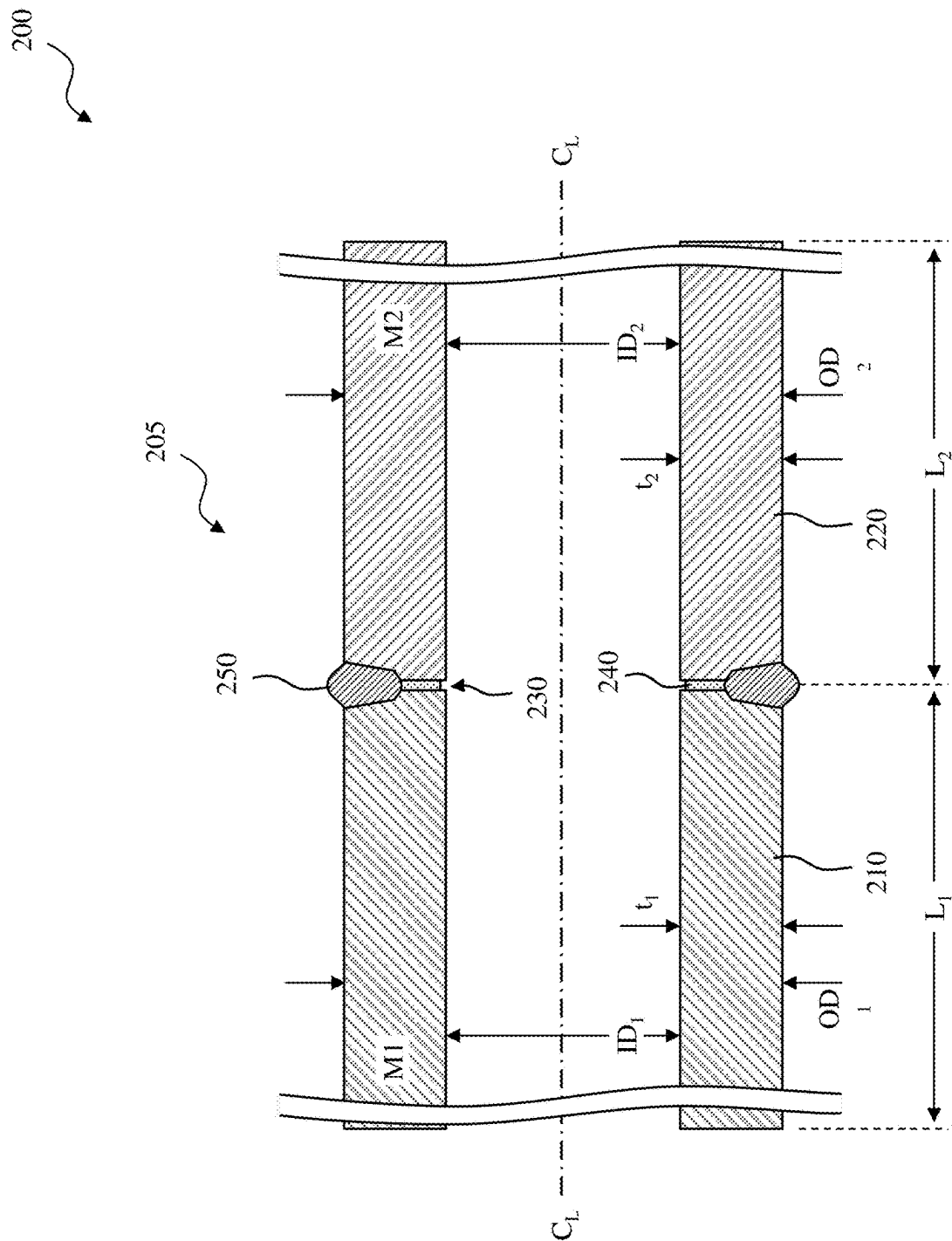


FIG. 2

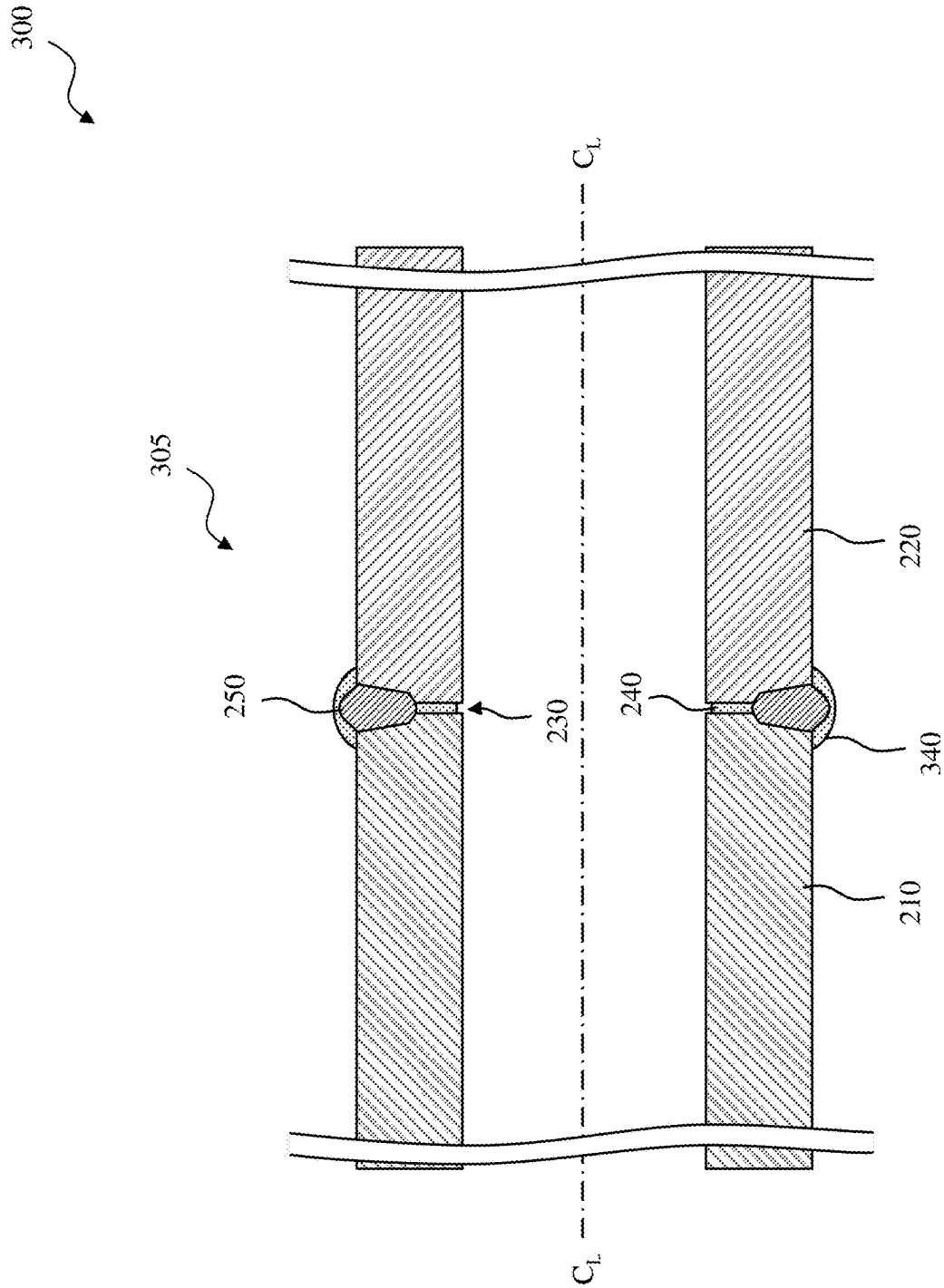


FIG. 3

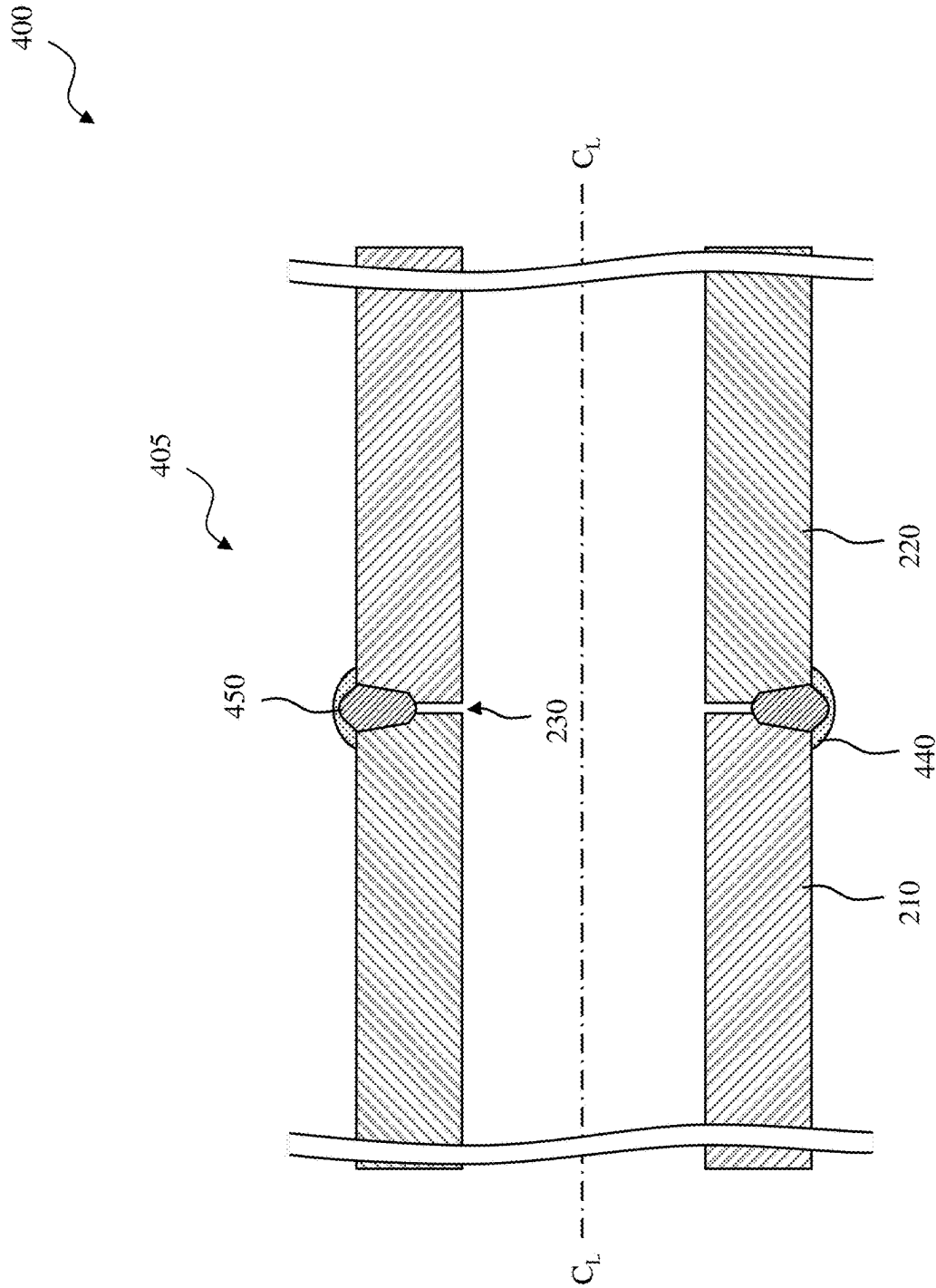


FIG. 4

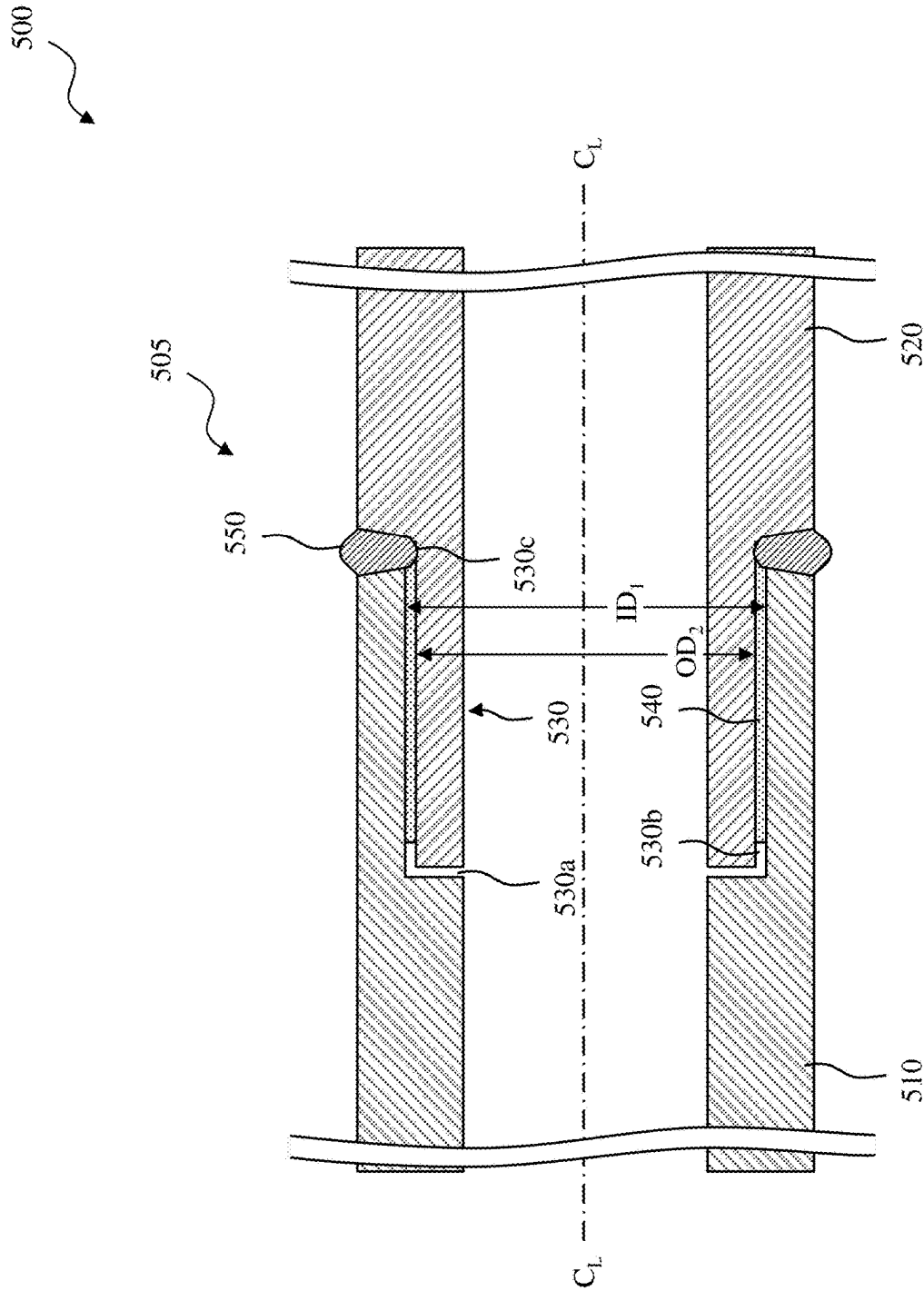


FIG. 5

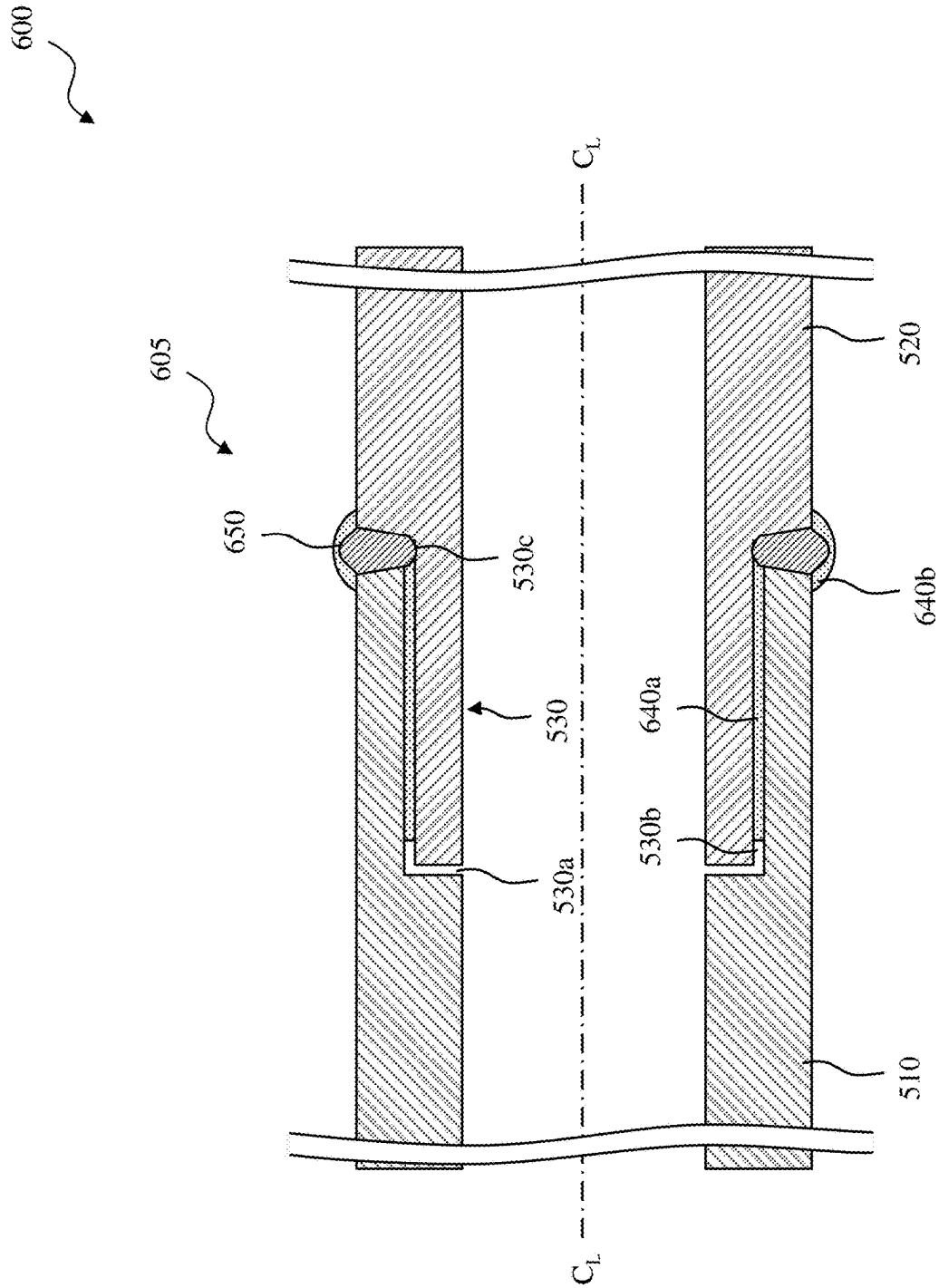


FIG. 6

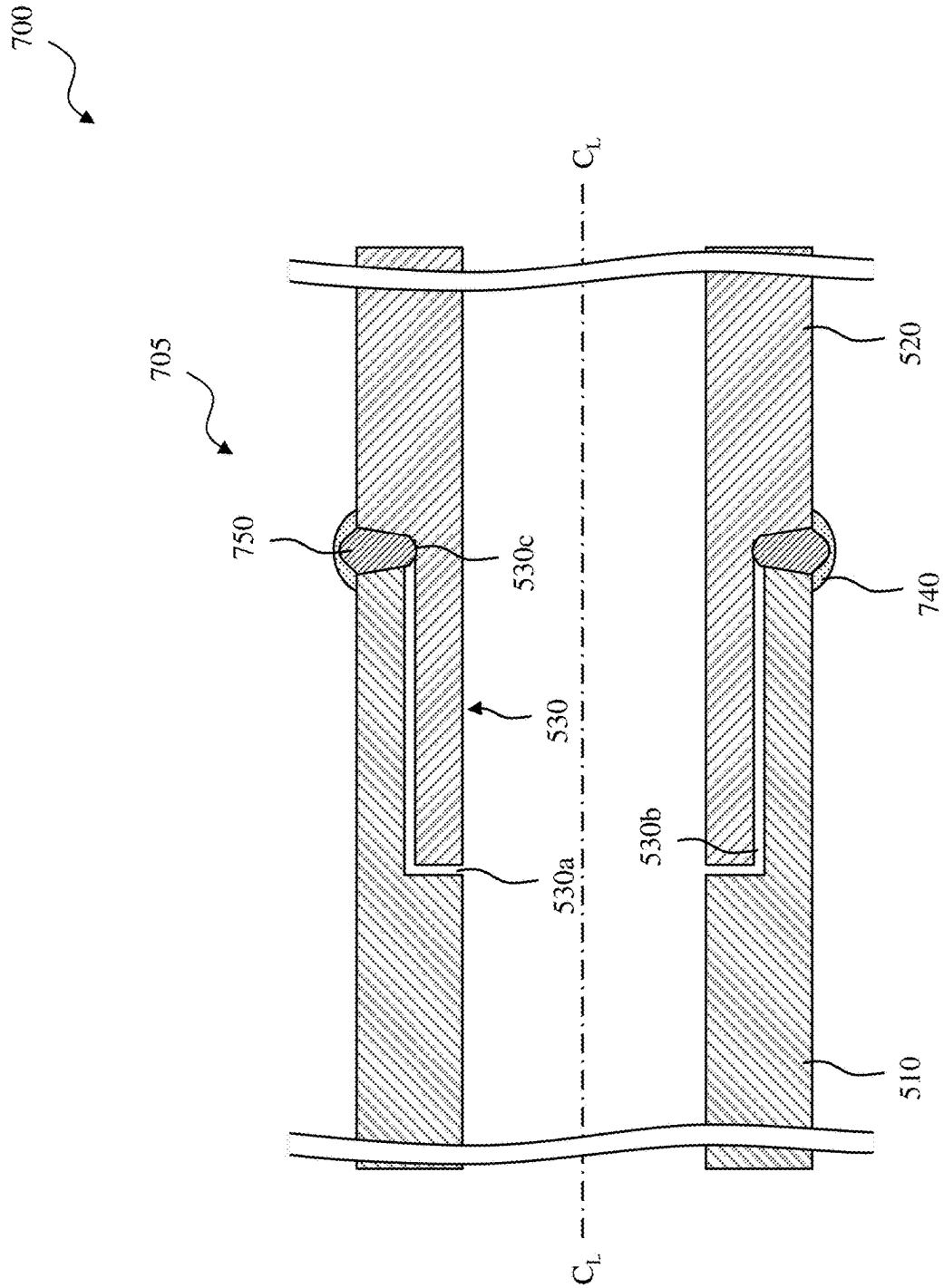


FIG. 7

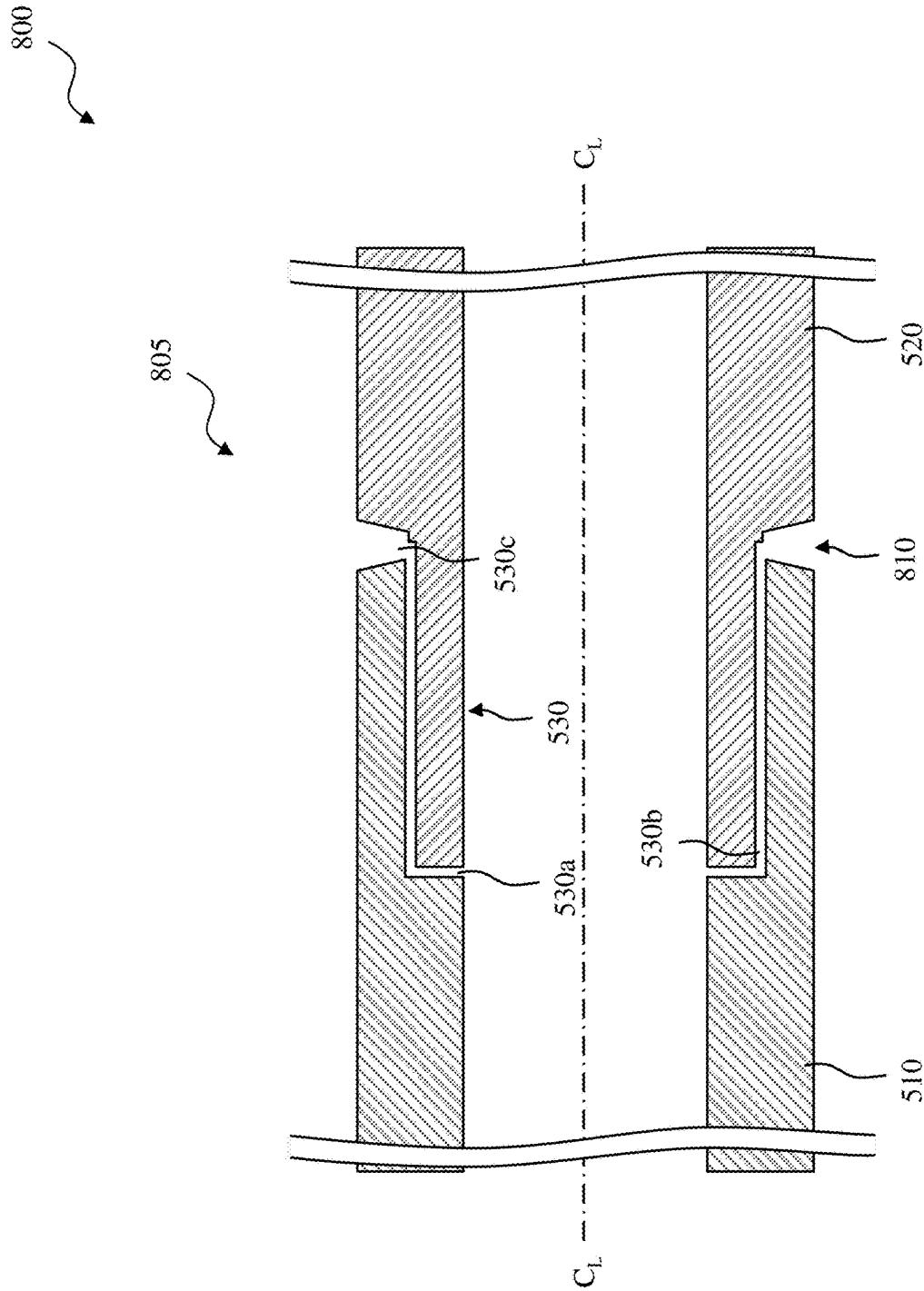


FIG. 8A

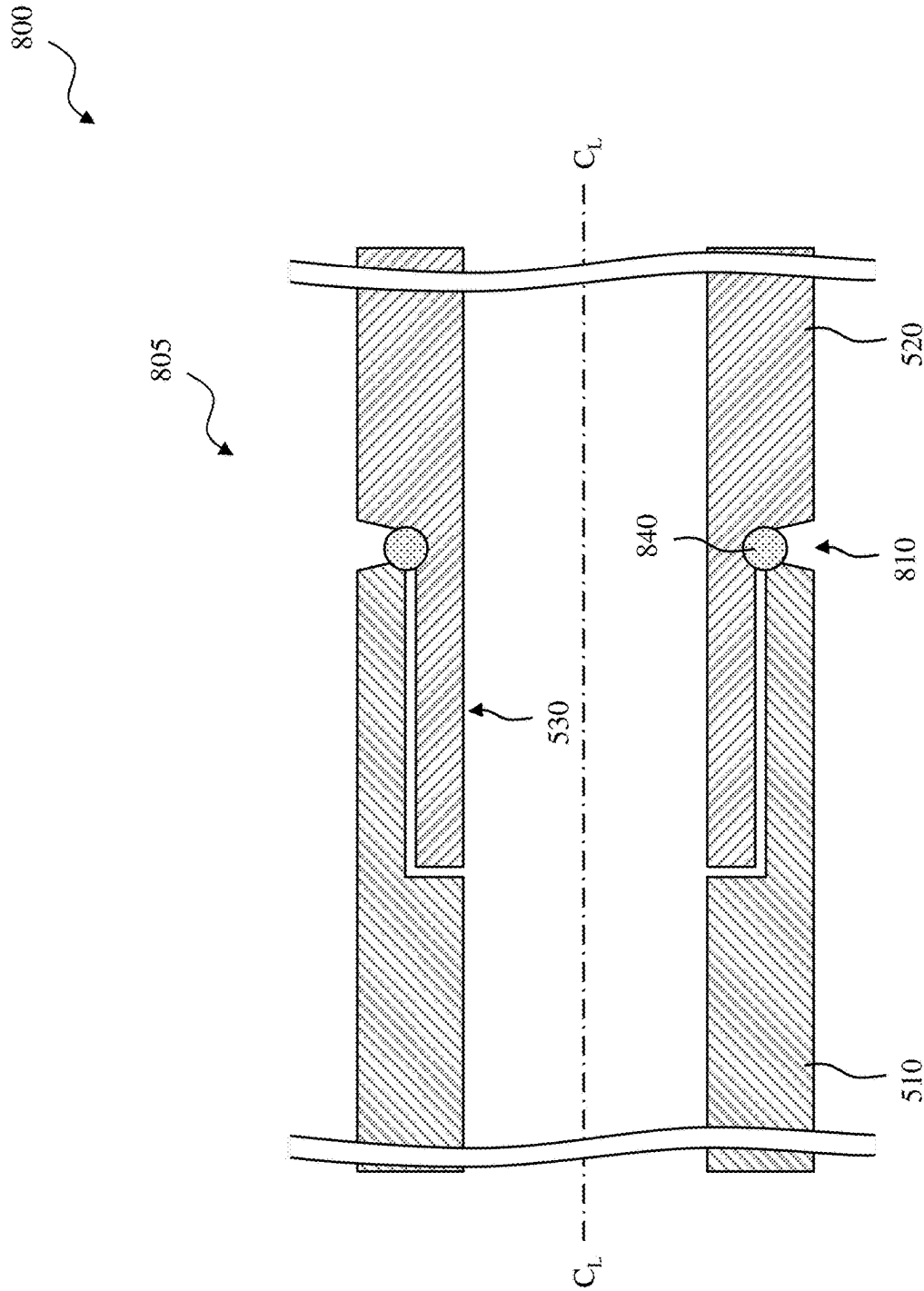


FIG. 8B

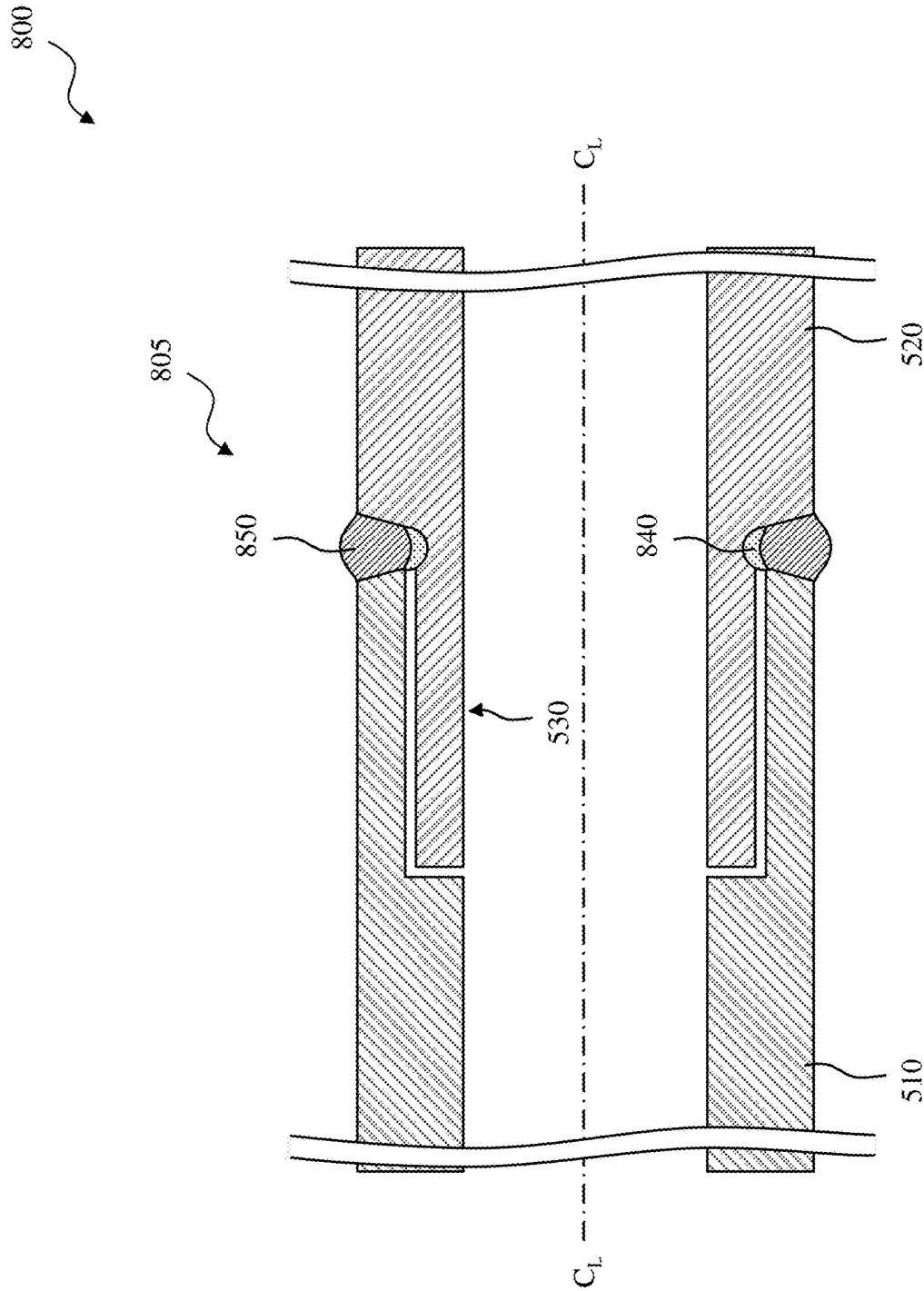


FIG. 8C

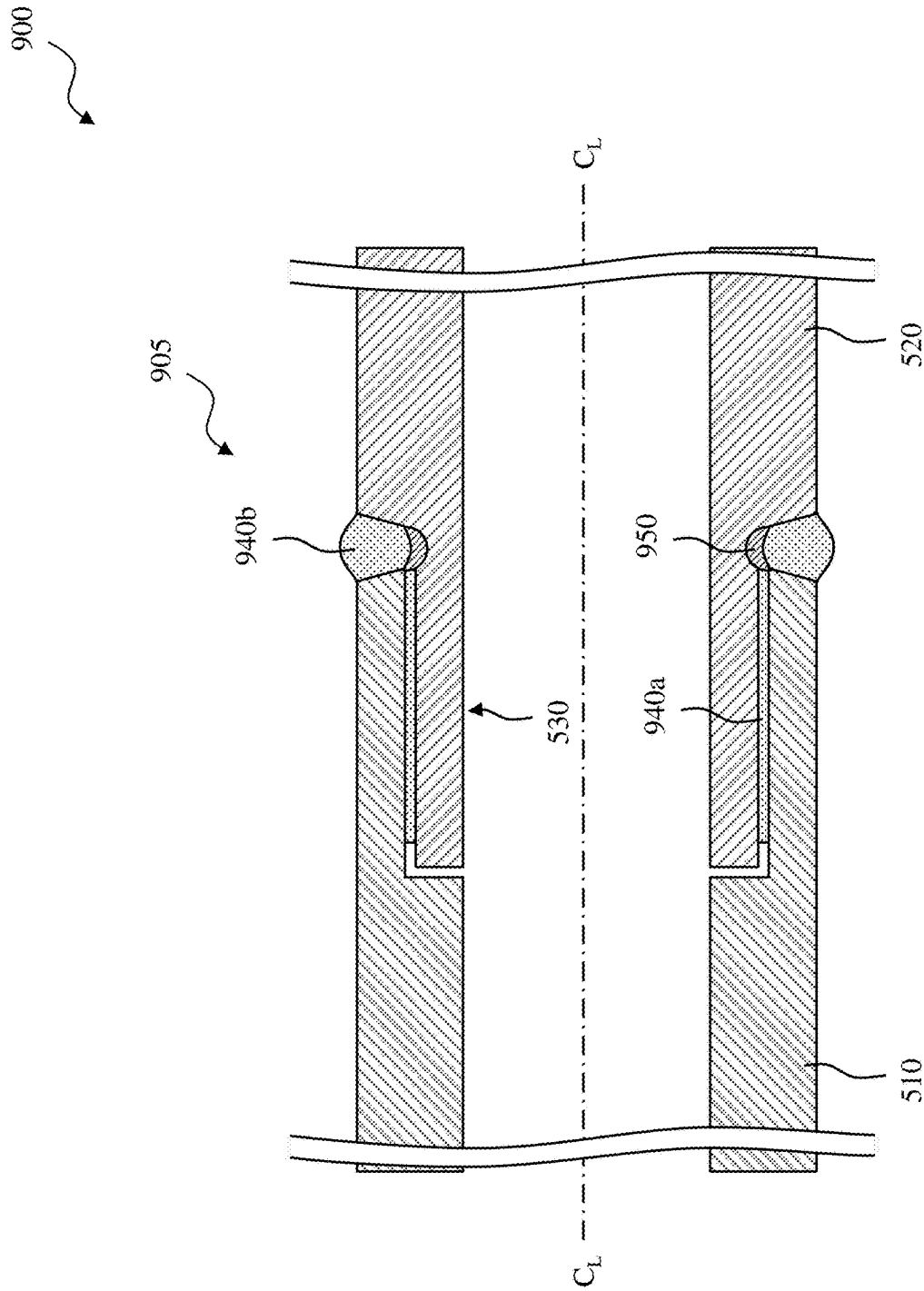


FIG. 9

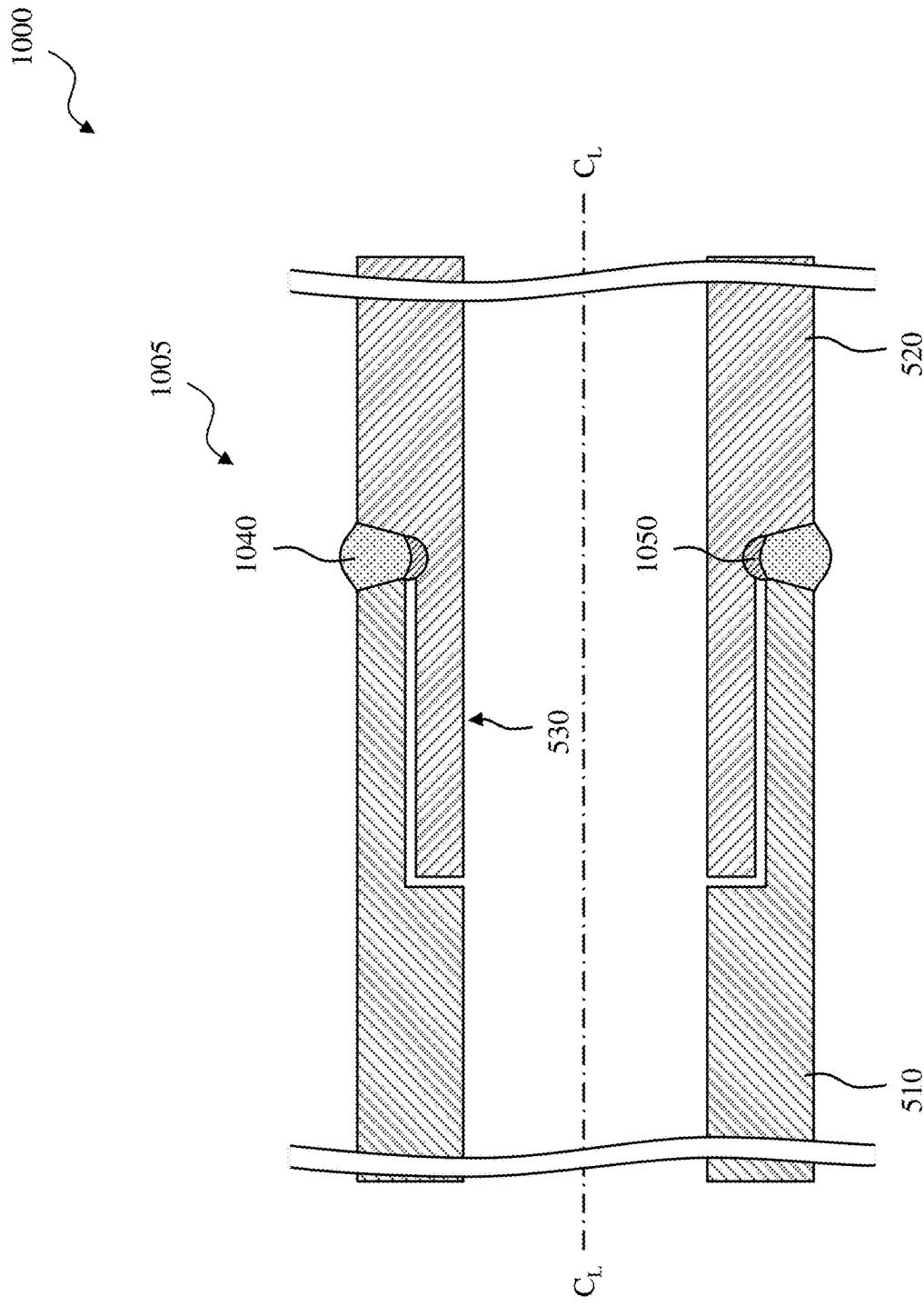


FIG. 10

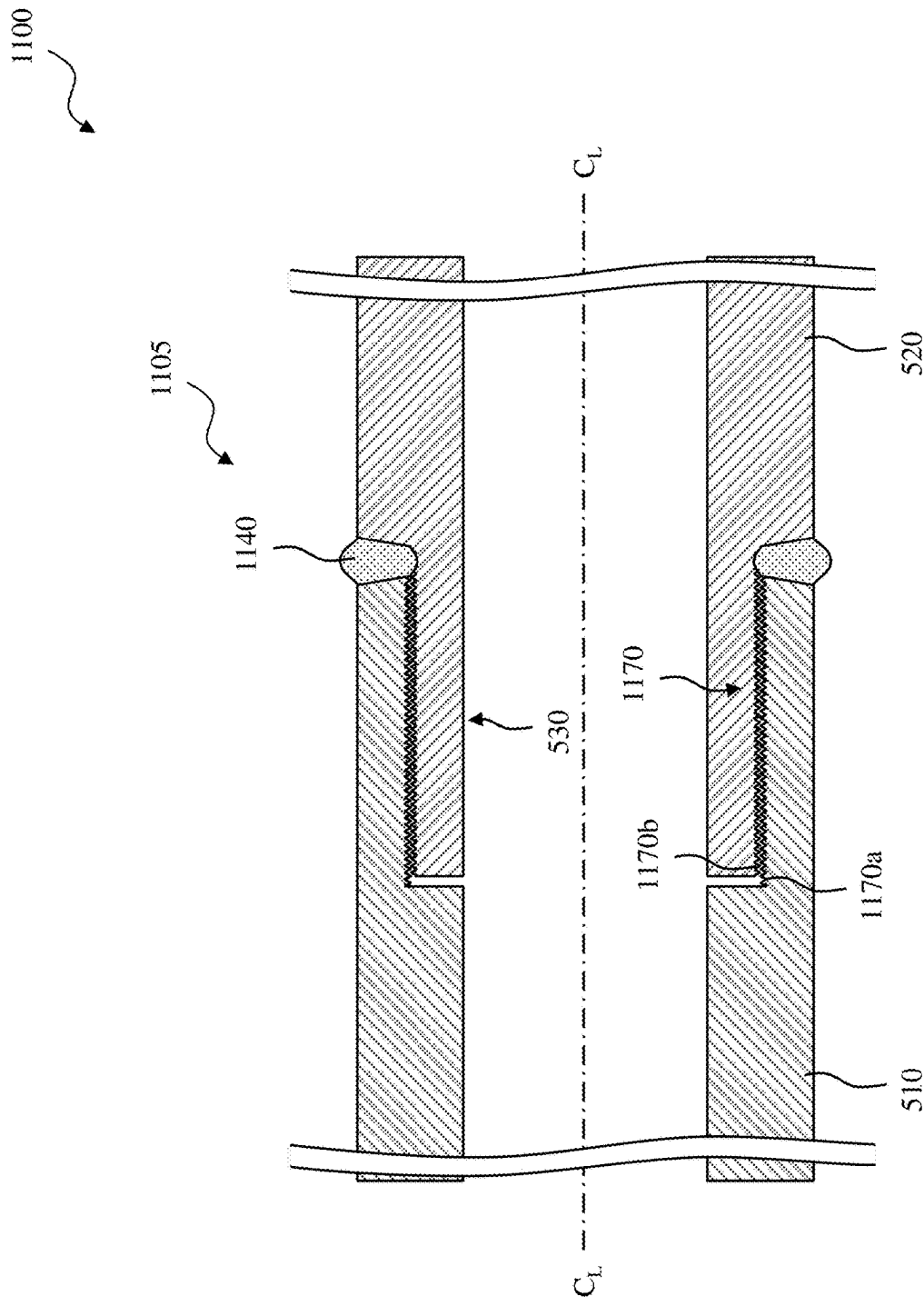


FIG. 11

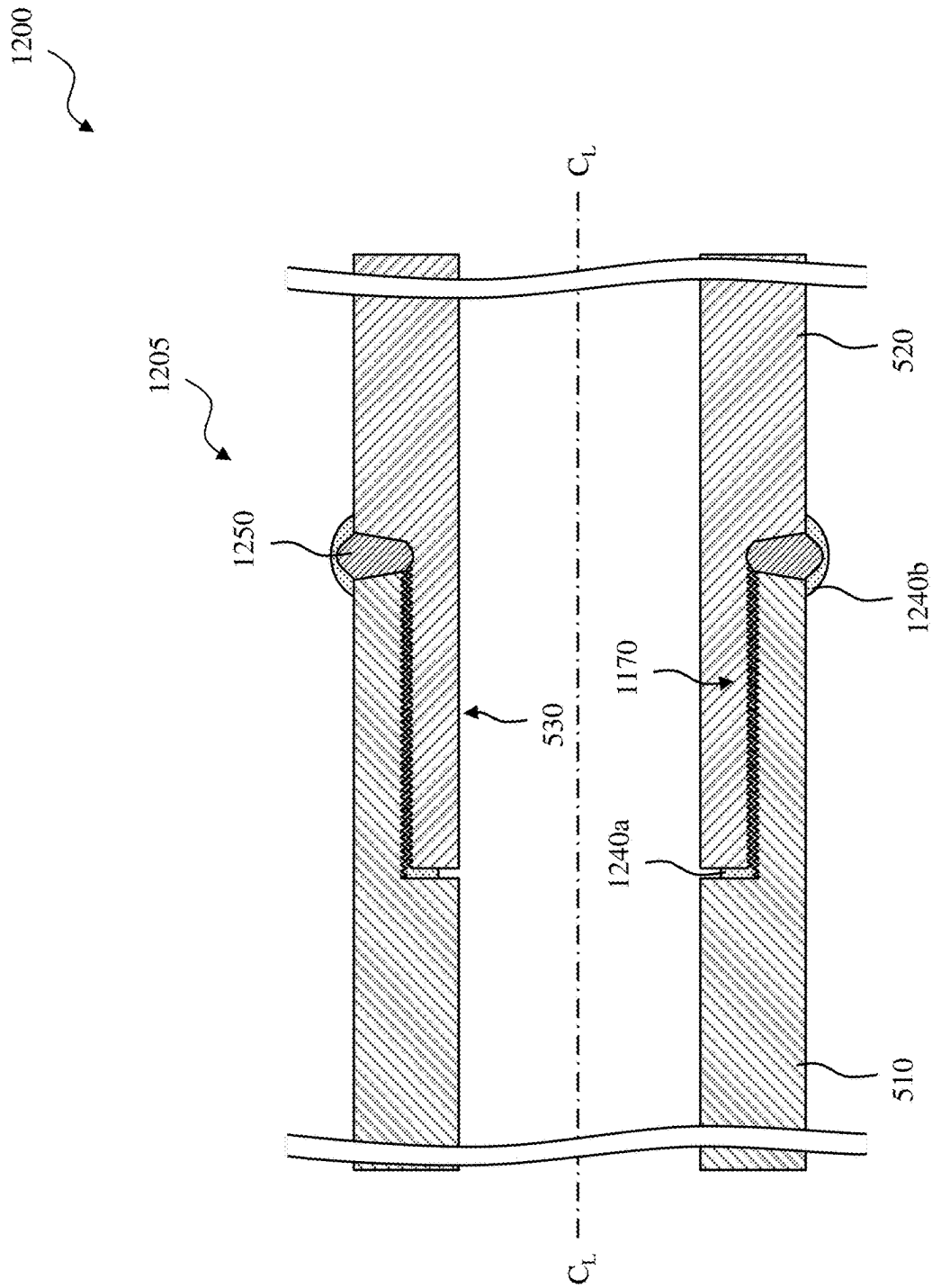


FIG. 12

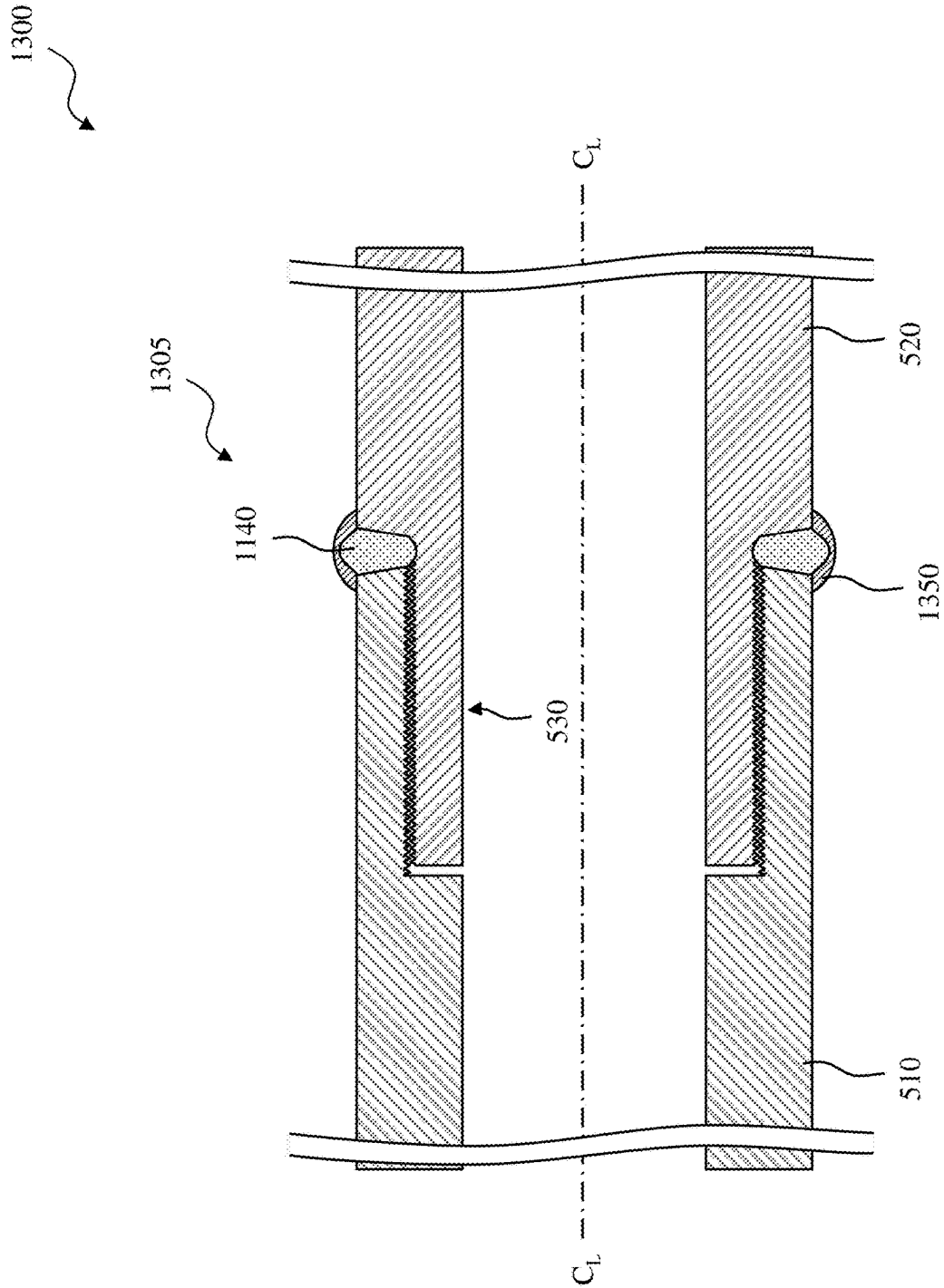


FIG. 13

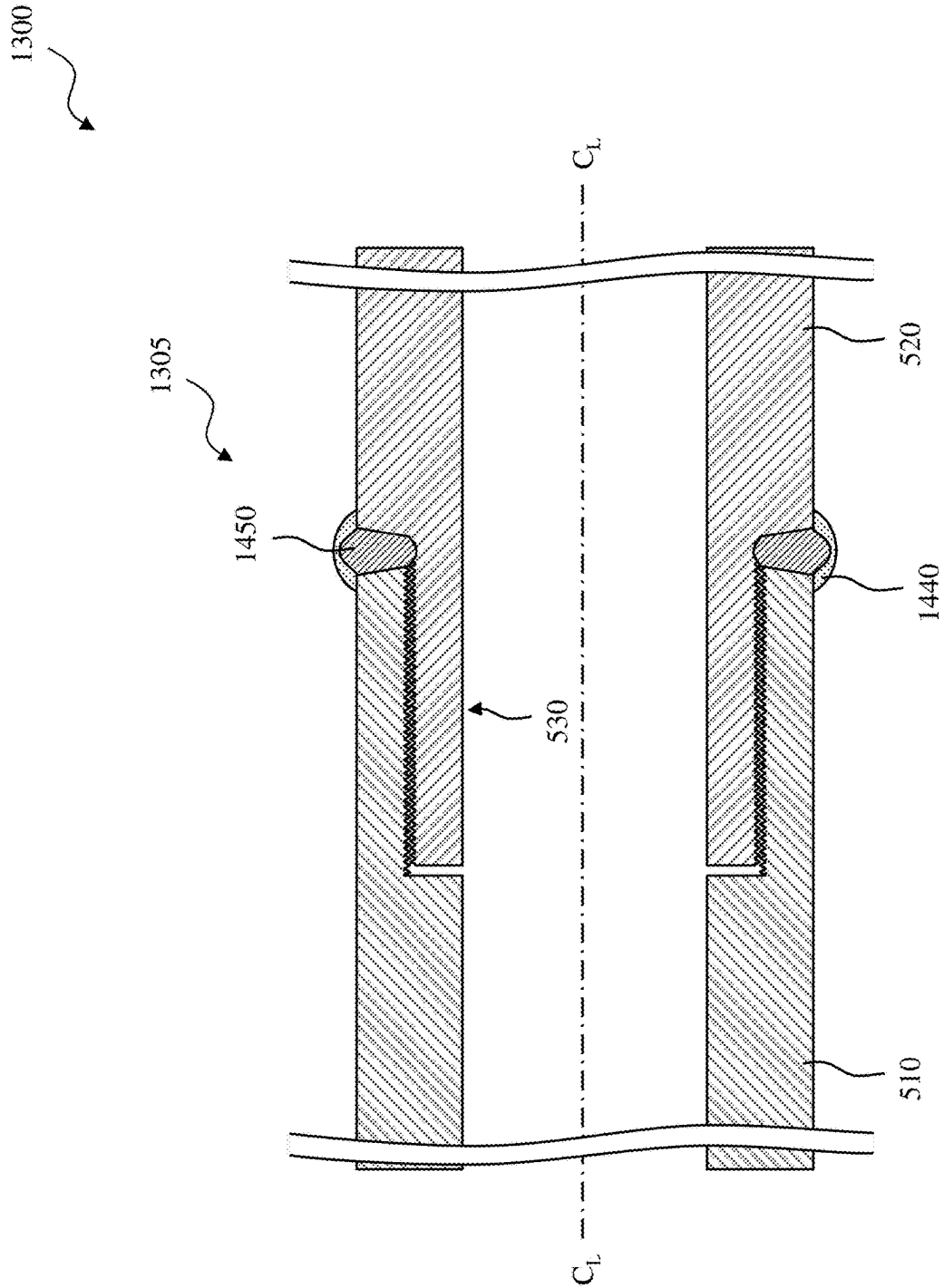


FIG. 14

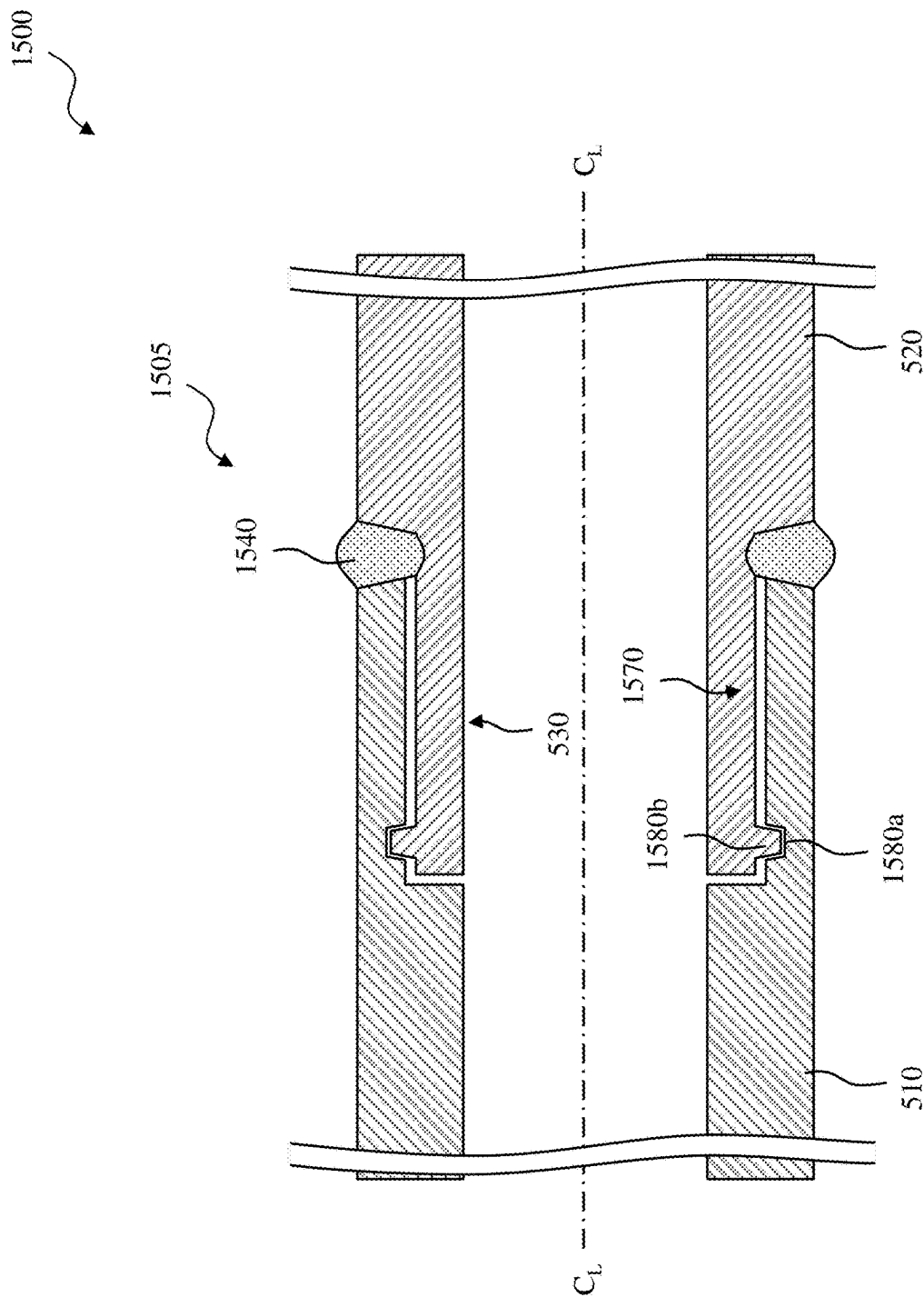


FIG. 15

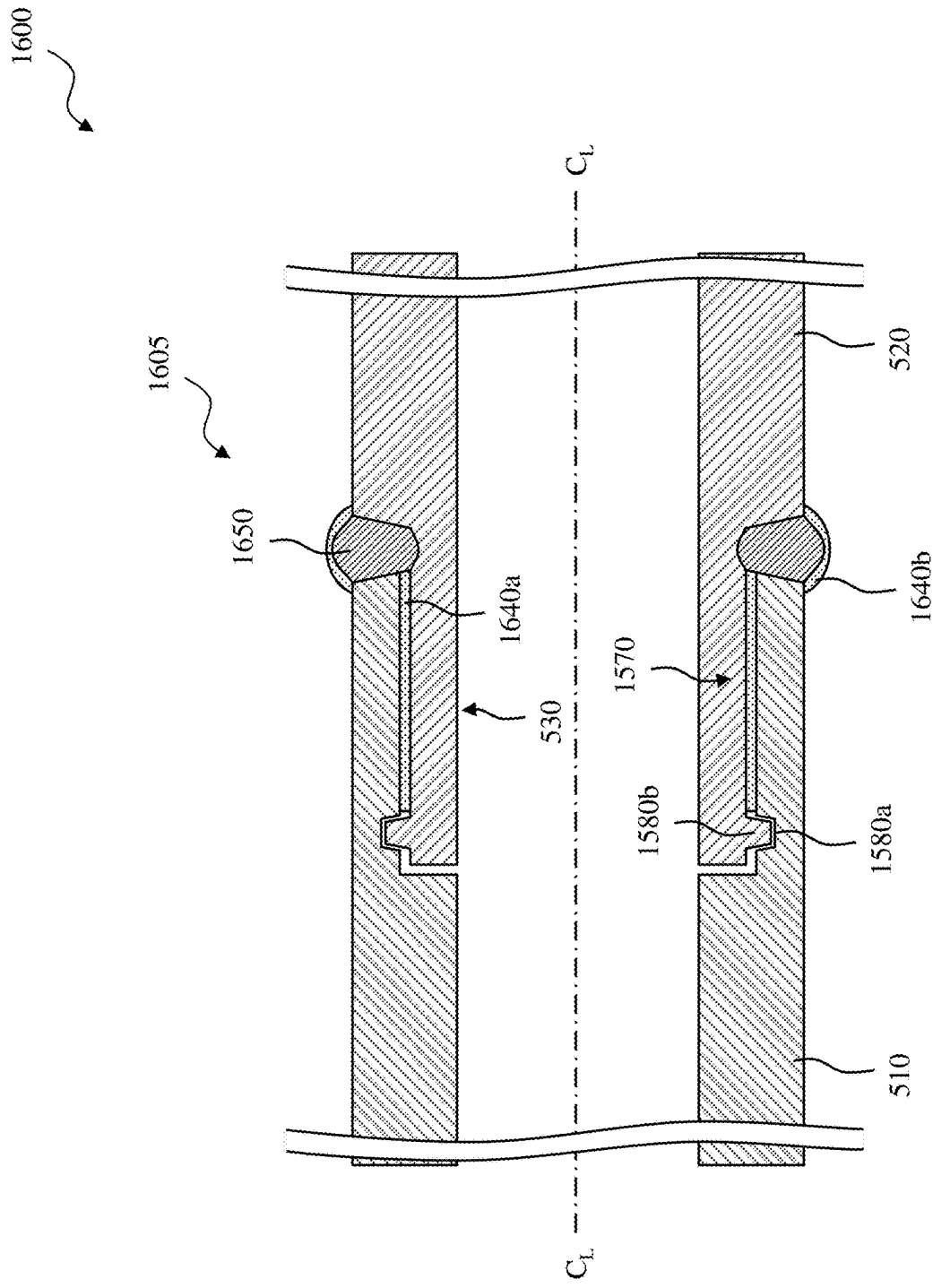


FIG. 16

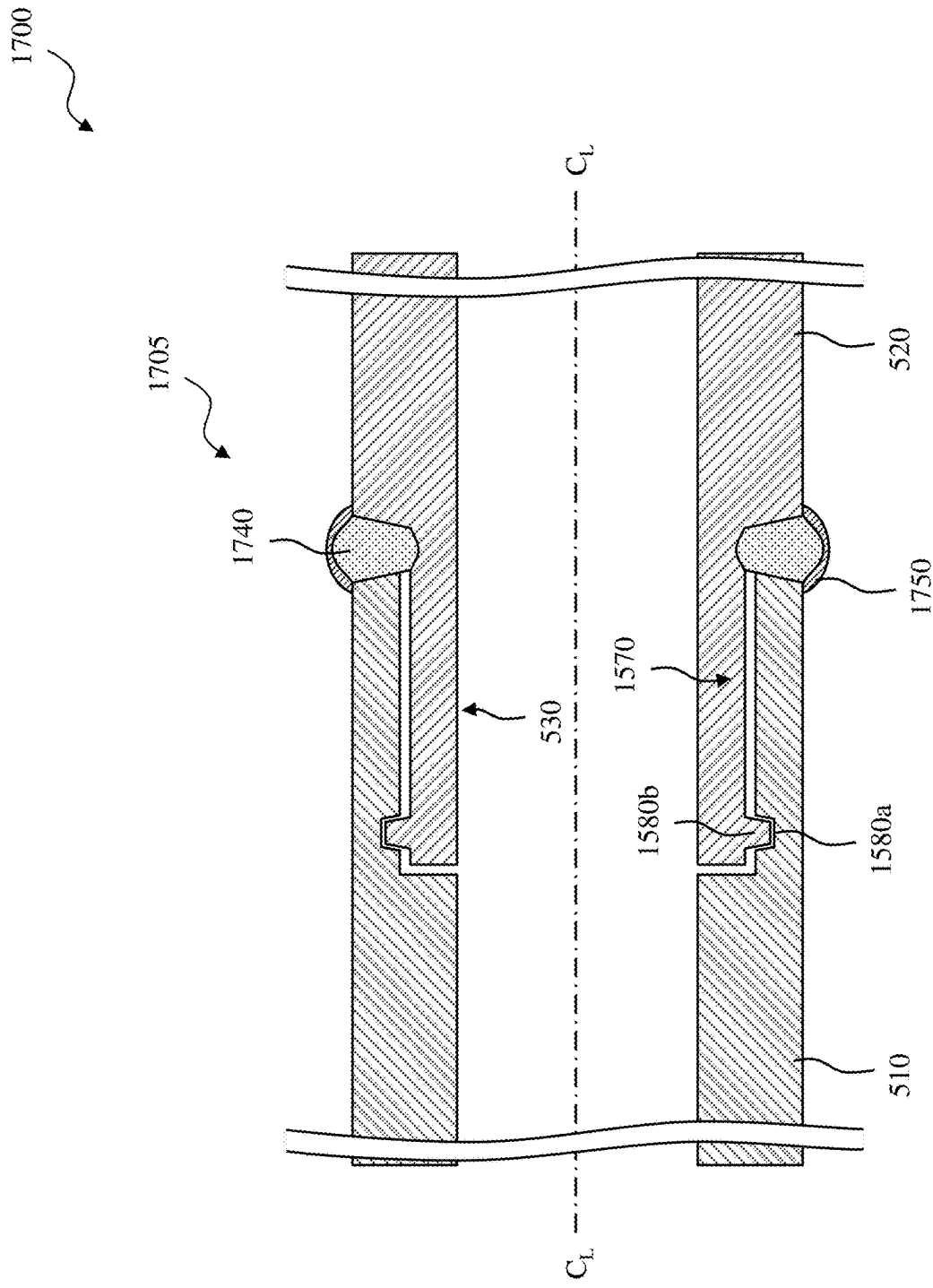


FIG. 17

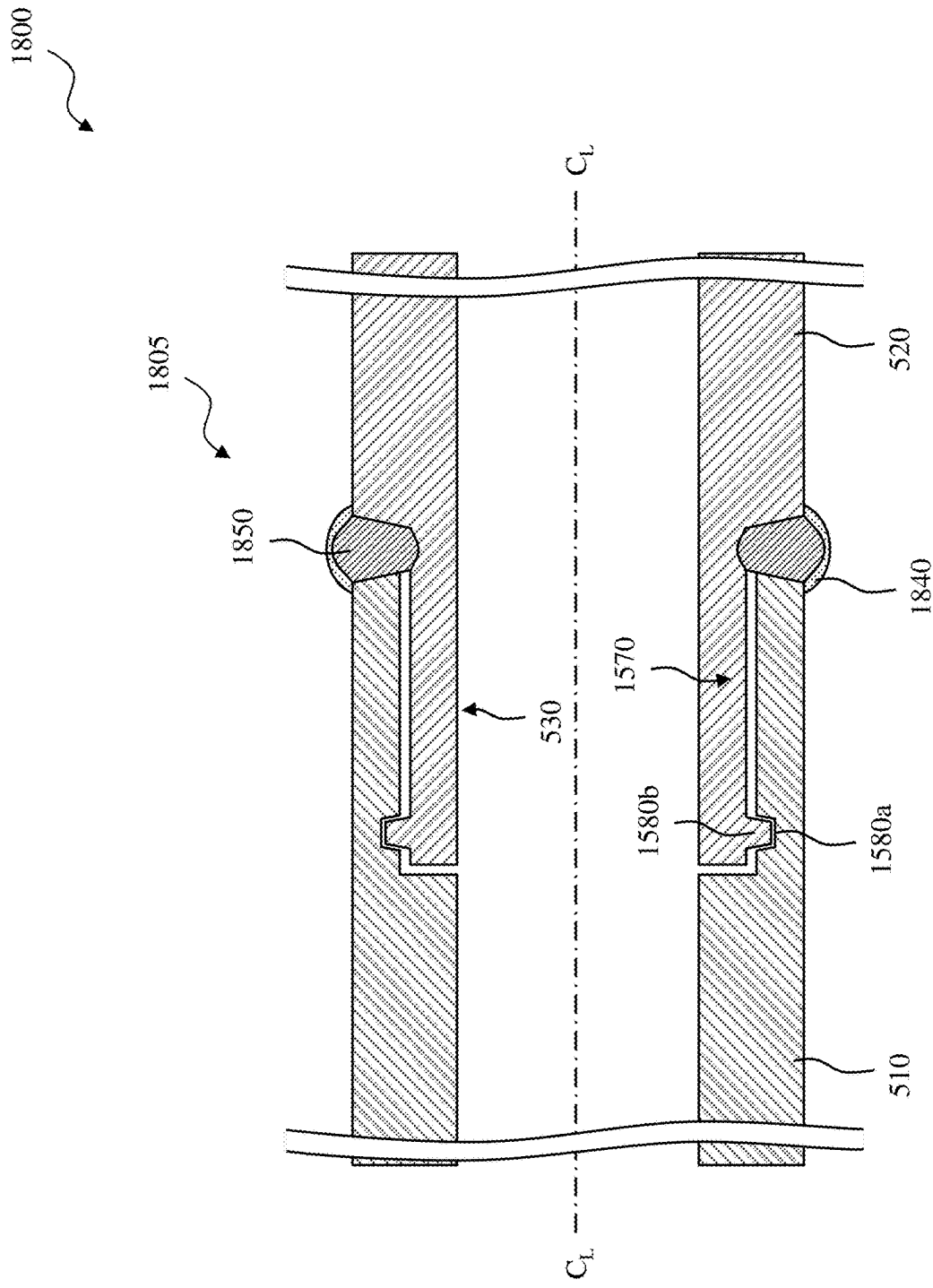


FIG. 18

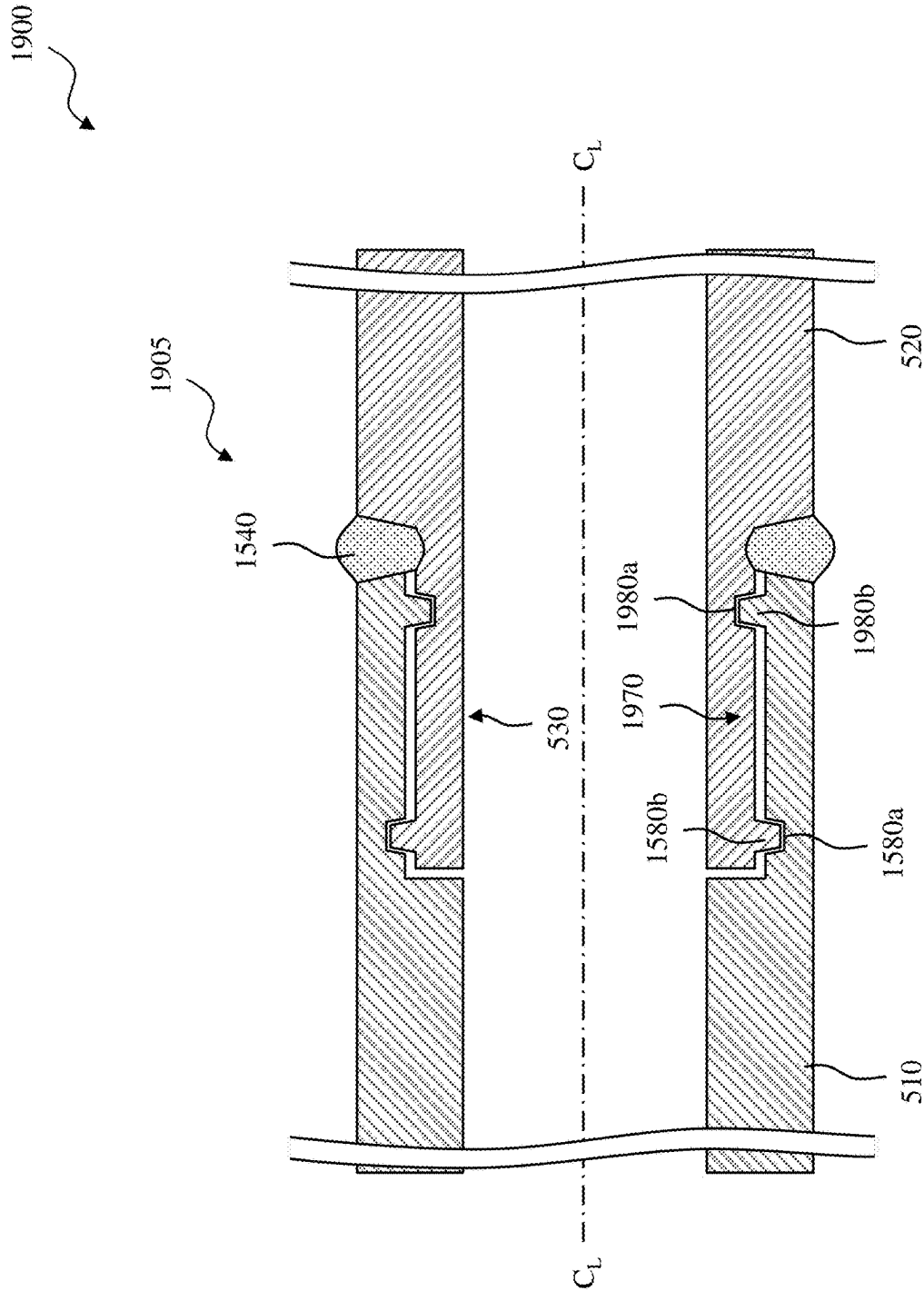


FIG. 19

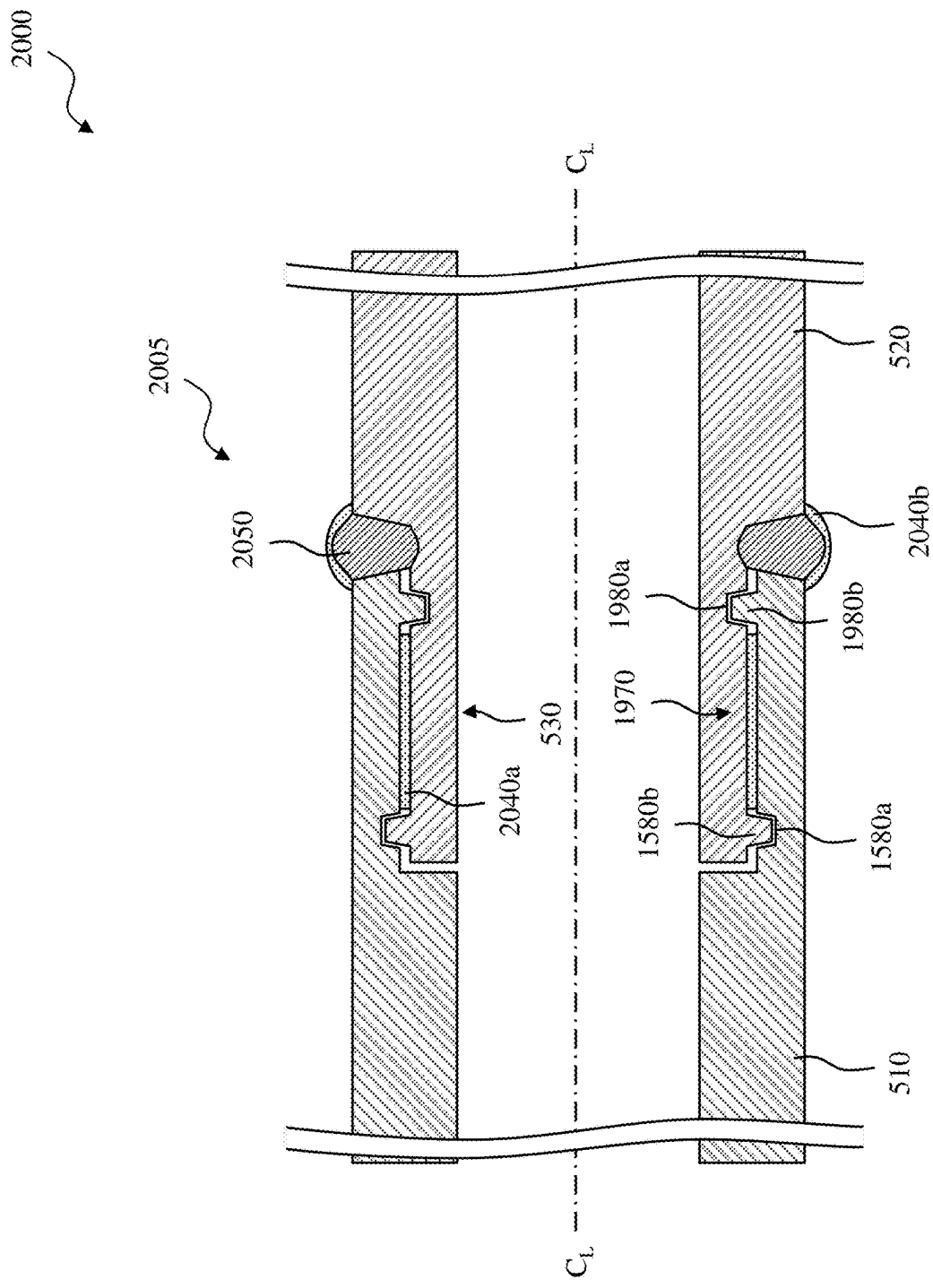


FIG. 20

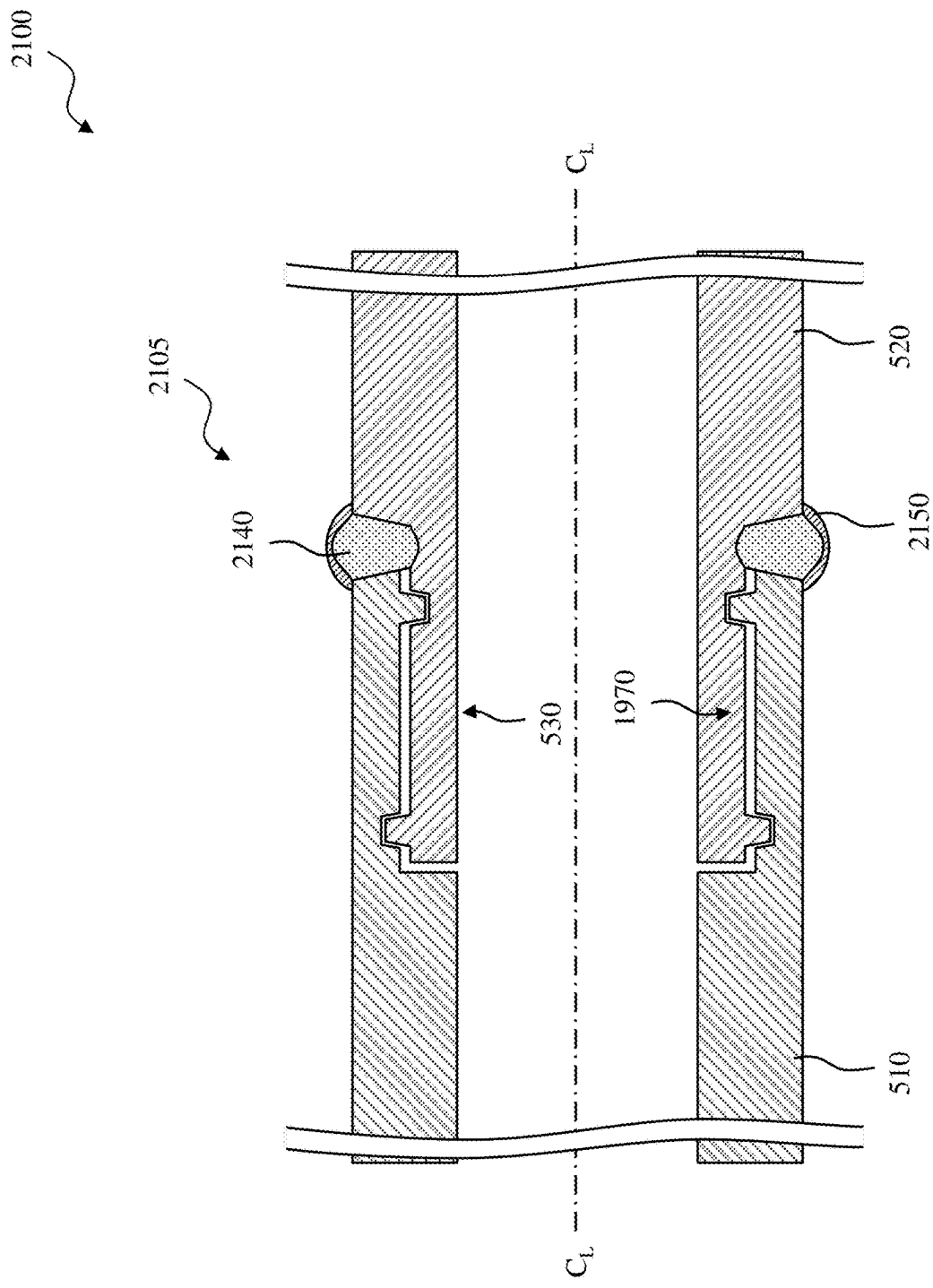


FIG. 21

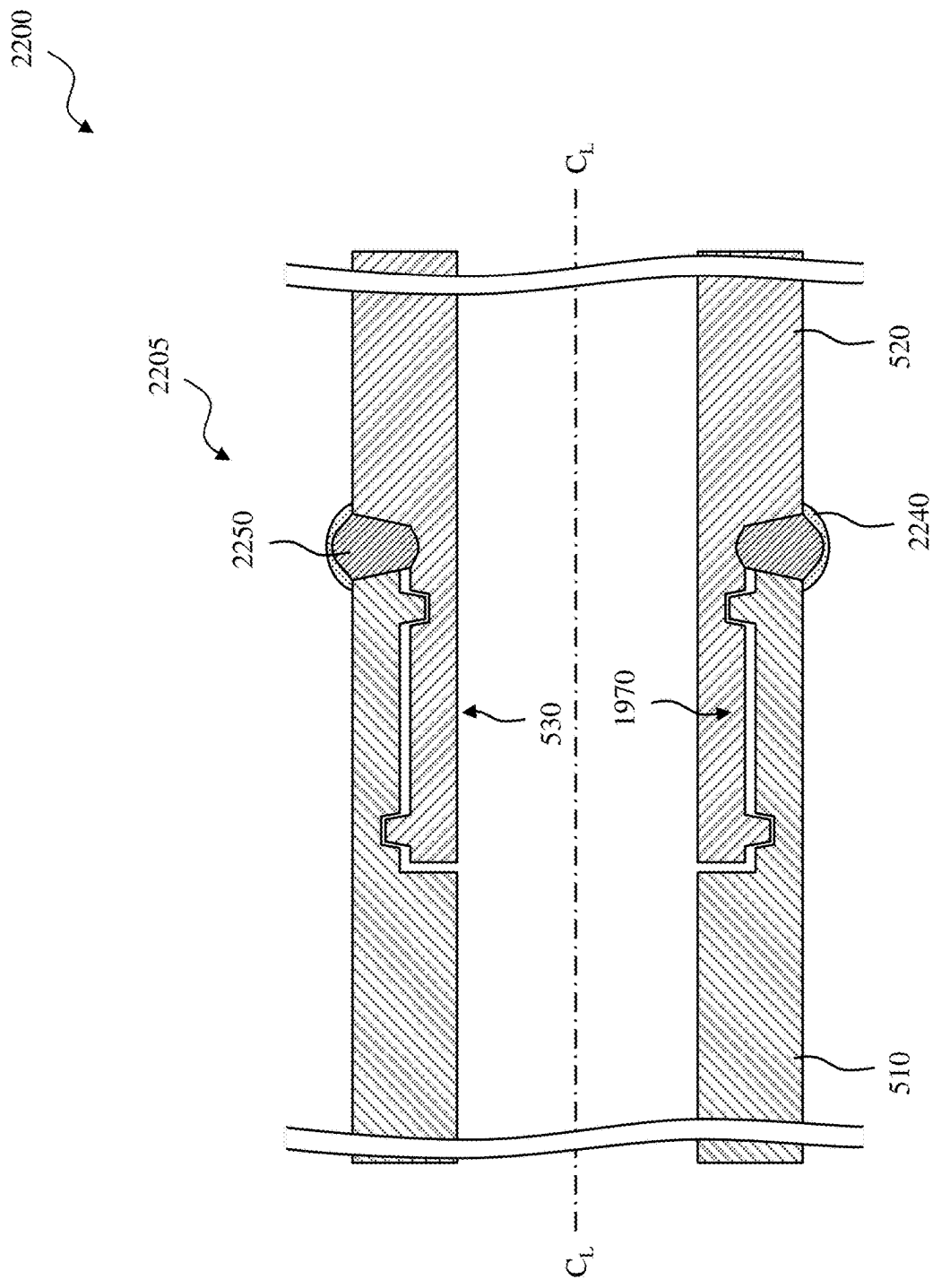


FIG. 22

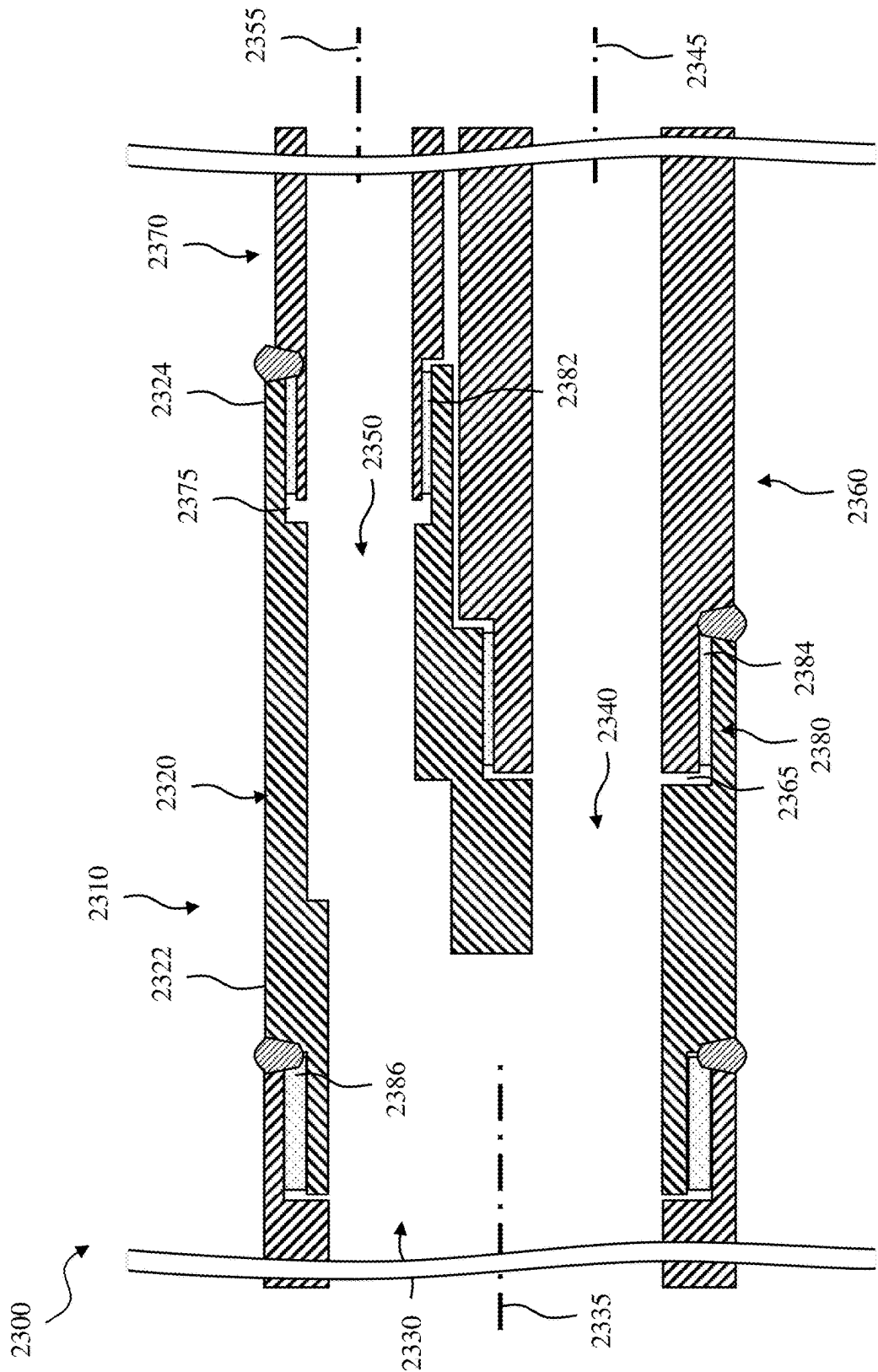


FIG. 23

**DOWNHOLE TOOL WITH CRACK
COMPLIANT SEAL AND MECHANICAL
STRENGTHENING FEATURE AT A JOINT
THEREOF**

**CROSS-REFERENCE TO RELATED
APPLICATION**

[0001] This application is a continuation of U.S. application Ser. No. 18/348,205, filed on Jul. 6, 2023, entitled “DOWNHOLE TOOL WITH CRACK COMPLIANT SEAL AND MECHANICAL STRENGTHENING FEATURE AT A JOINT THEREOF,” which claims the benefit of U.S. Provisional Application Ser. No. 63/511,392, filed on Jun. 30, 2023, entitled “DOWNHOLE TOOL WITH CRACK COMPLIANT SEAL AND HIGH YIELD STRENGTH WELD POSITIONED AT A JOINT THEREOF,” commonly assigned with this application and incorporated herein by reference in their entirety.

BACKGROUND

[0002] Traditional tubular joints that perform simultaneous anchoring and sealing between two different parts may be achieved by using a combination of geometric mechanical joining methods, and sealing elements or inserts (e.g., elastomeric/plastic/metal). For example, geometric mechanical joining methods including non-sealing threads, snap rings, collets, Ratch Latch™, lock rings, bolting/riveting and other type of latching methods are often used.

BRIEF DESCRIPTION

[0003] Reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

[0004] FIG. 1 illustrates a well system designed, manufactured, and operated according to one or more embodiments of the disclosure, and including a multilateral junction (e.g., y-block and two or more wellbore legs), and/or an interval control valve (ICV), and/or downhole packer, designed, manufactured and operated according to one or more embodiments of the disclosure; and

[0005] FIGS. 2 through 23 illustrate various different embodiments for downhole tools including improved joints designed, manufactured and operated according to one or more embodiments of the disclosure.

DETAILED DESCRIPTION

[0006] In the drawings and descriptions that follow, like parts are typically marked throughout the specification and drawings with the same reference numerals, respectively. The drawn figures are not necessarily to scale. Certain features of the disclosure may be shown exaggerated in scale or in somewhat schematic form and some details of certain elements may not be shown in the interest of clarity and conciseness. The present disclosure may be implemented in embodiments of different forms.

[0007] Specific embodiments are described in detail and are shown in the drawings, with the understanding that the present disclosure is to be considered an exemplification of the principles of the disclosure, and is not intended to limit the disclosure to that illustrated and described herein. It is to be fully recognized that the different teachings of the embodiments discussed herein may be employed separately or in any suitable combination to produce desired results.

[0008] Unless otherwise specified, use of the terms “connect,” “engage,” “couple,” “attach,” or any other like term describing an interaction between elements is not meant to limit the interaction to direct interaction between the elements and may also include indirect interaction between the elements described.

[0009] Unless otherwise specified, use of the terms “up,” “upper,” “upward,” “uphole,” “upstream,” or other like terms shall be construed as generally toward the surface of the ground; likewise, use of the terms “down,” “lower,” “downward,” “downhole,” or other like terms shall be construed as generally toward the bottom, terminal end of a well, regardless of the wellbore orientation. Use of any one or more of the foregoing terms shall not be construed as denoting positions along a perfectly vertical axis. Unless otherwise specified, use of the term “subterranean formation” shall be construed as encompassing both areas below exposed earth and areas below earth covered by water such as ocean or fresh water.

[0010] The present disclosure is based, at least in part, on the recognition that there is an ever increasing need to design and/or develop downhole tools capable of handling greater collapse and/or burst pressures. Additionally, the newly developed downhole tools need to be corrosion resistant, or in certain instances at a minimum National Association of Corrosion Engineers (NACE) compliant, and/or International Organization for Standardization (ISO) compliant. Specifically, the downhole tools, and any joints that they may have, may need to be corrosion resistant to hydrogen sulfide (H₂S) gas, but in certain embodiments corrosion resistant to carbon dioxide (CO₂) gas and/or hydrogen (H₂) gas. Unfortunately, as recognized in the present disclosure, high yield strength joints tend to lack the necessary NACE/ISO compliance, and NACE/ISO compliant joints tend to lack the necessary yield strength.

[0011] Other environments which may comprise corrosive fluids (hydrocarbons, H₂S fluids, CO₂ fluids, acids, bases, etc.), contaminants (sand, debris, paraffins, asphaltenes, etc.), high-temperature fluids (fluids from geothermal formations, injected fluids, etc.), cryogenic fluids, etc. may benefit from this disclosure as well. Moreover, the ability to provide systems and methods for making and utilizing tubular joints capable of providing both high yield strength and crack compliance for use in harsh conditions (e.g., corrosive environments or contaminated fluids), extreme pressures (e.g., >5,000-psi differential), extreme temperatures (e.g., >-20° F. or >300° F.), and/or in remote locations, makes this disclosure suitable for use in other remote locations with harsh environments such as outer space (e.g., satellites, spacecrafts, etc.), aeronautics (aircrafts, drones), on-ground (swamps, marshes, power generation, hydrogen or other gas extraction and/or transportation, etc.), below ground (mines, caves, etc.), ocean (on surface and subsea), subterranean (mineral extraction, storage wells (carbon sequestration, carbon capture and storage (CCS), etc.)), and other energy recovery activities (geothermal, steam, etc.).

[0012] With these recognitions in mind, the present disclosure has developed one or more different designs for tubular joints providing both crack compliance (e.g., NACE/ISO compliance) and high yield strength. In at least one embodiment, the downhole tool includes first and second members positioned proximate one another to form an overlapping space therebetween, and then both a crack compliant seal (e.g., crack compliant compound, solder,

braze, weld, etc.) and a high yield strength weld are positioned at the joint, the high yield strength weld having a yield strength of at least 50 kilopound per square inch (ksi). In such a circumstance, a joint with both crack compliance and a yield strength of at least 50 ksi may be obtained. In at least one other embodiment, the high yield strength weld may have a yield strength of at least 80 ksi, which may provide a joint with both crack compliance and a yield strength of at least 80 ksi. In at least one other embodiment, first and second members are positioned proximate one another to form an overlapping space therebetween, and a geometric mechanical strengthening feature (e.g., configured to increase an engineering rating of the overlapping space, such as the burst rating and/or collapse rating and/or tensile rating and/or compression rating and/or any combination thereof) and the crack compliant seal (e.g., crack compliant compound, solder, braze, and/or weld, etc.) are used. In such a circumstance, a mechanically strengthened joint with both crack compliance and a yield strength of at least 50 ksi (e.g., or even 80 ksi) may also be obtained.

[0013] As will be understood more fully below, the crack compliant seal may comprise many different materials and/or structures/locations and remain within the scope of the present disclosure. In at least one embodiment, the crack compliant seal has a compliant seal hardness value (HV_{cs}) less than or equal to one or more of the first member hardness value (HV_1) or second member hardness value (HV_2). The hardness values (HV) may be measured using the Vickers hardness test. Accordingly, the hardness values (HV) may be determined with the ratio F/Q , where F is the force applied to a diamond in the form of a square-based pyramid in kilograms-force (kgf), and A is the surface area of the resulting indentation in square millimeters (mm). In at least one embodiment, A can be determined by the formula: $A = d^2/2 \sin(136^\circ/2)$, which can be approximated by evaluating the sine term to give, $A \approx d^2/1.8544$, where d is the average length of the diagonal left by the indenter in millimeters. Thus, in at least one embodiment the hardness value (HV) is approximately equal to $1.8544F/d^2$, where F is in kgf and d is in millimeters.

[0014] Further to at least one embodiment, the compliant seal hardness value (HV_{cs}) is a post weld heat treatment compliant seal hardness value (HV_{cs-pw}). In at least one other embodiment, the post weld compliant seal hardness value (HV_{cs-pw}) is 250 or less. In yet another embodiment, the post weld compliant seal hardness value (HV_{cs-pw}) is 250 or less after being subjected to the post weld heat treatment of at least 620° C. In even another embodiment, the crack compliant seal includes no more than 1% mass fraction of nickel. In even another embodiment, the compliant seal comprises (e.g., in whole or in part) a non-metallic material, such as polyether ether ketone (PEEK), fluorocarbon rubber (VITON®), a polymer, a synthetic polymer (e.g., hydrogenated nitrile “HNBR”), etc.

[0015] In even yet another embodiment, the crack compliant seal is a NACE/ISO compliant seal. For example, the NACE/ISO compliant seal, which would form a NACE/ISO compliant junction with the first and second members, would be selected from the NACE/ISO compliant seals, junctions, and/or first and second members, as set forth in ANSI/NACE MR0175/ISO (15156-1:2015), entitled “Petroleum, petrochemical, and natural gas industries—Materials for use in H_2S -containing environments in oil and gas production,” the entirety of which is incorporated herein by

reference, as if it were fully reproduced herein. In the event of inconsistent usages between this disclosure and the ANSI/NACE MR0175/ISO (15156-1:2015) so incorporated by reference, the usage in ANSI/NACE MR0175/ISO (15156-1:2015) should be considered supplementary to that of this disclosure; for irreconcilable inconsistencies, the usage in this disclosure controls.

[0016] In at least one embodiment, the NACE/ISO compliant seal has a pitting-resistance equivalent number (PREN or F_{PREN}) of at least 30. In yet another embodiment, the F_{PREN} is at least 35, if not somewhere between 30 and 40. In yet another embodiment, the F_{PREN} is at least 40, if not somewhere between 35 and 45. In yet another embodiment, the F_{PREN} is at least 40, if not somewhere between 40 and 45. In these embodiments, the PREN (F_{PREN}) shall be calculated as given in Formula (1):

$$F_{PREN-WCr} = [3, 3 \dots w_{Mo} 0, 5w_W \dots] 16w_N \quad (1)$$

where

[0017] w_{Cr} is the mass fraction of chromium in the alloy, expressed as a percentage mass fraction of the total composition;

[0018] w_{Mo} is the mass fraction of molybdenum in the alloy, expressed as a percentage mass fraction of the total composition;

[0019] w_W is the mass fraction of tungsten in the alloy, expressed as a percentage mass fraction of the total composition; and

[0020] w_N is the mass fraction of nitrogen in the alloy, expressed as a percentage mass fraction of the total composition.

[0021] In at least one other embodiment, the high yield strength weld is a very high yield strength weld having a yield strength of at least 95 ksi, a super high yield strength weld having a yield strength of at least 110 ksi, of an extremely high yield strength weld having a yield strength of at least 125 ksi, or an excessively high yield strength weld having a yield strength of at least 140 ksi.

[0022] The term “yield strength,” as used herein, is the stress at which a material exhibits a specified deviation from the proportionality of stress to strain. The term “braze,” as used herein, means joining metals by flowing a thin layer (e.g., of capillary thickness) of a lower-melting-point non-ferrous filler. The term “weld,” as used herein, means the joining of two or more pieces of metal by applying heat and/or pressure with or without filler metal, to produce a union through localized fusion of the substrates and solidification across the interfaces.

[0023] FIG. 1 illustrates a well system 100 designed, manufactured, and operated according to one or more embodiments of the disclosure, and including a multilateral junction 175 (e.g., y-block and two or more wellbore legs) and/or interval control valve (ICV) 180 designed, manufactured and operated according to one or more embodiments of the disclosure. In accordance with at least one embodiment, the multilateral junction 175 and/or ICV 180 could include joints as disclosed herein. The well system 100 additionally includes one or more packers 185 designed, manufactured and operated according to one or more embodiments of the disclosure. In accordance with at least one embodiment, the one or more packers 185 could include

joints as disclosed herein. While not illustrated, the joints of the present disclosure may be used to join any two members employed downhole within the well system **100**.

[0024] The well system **100** includes a platform **120** positioned over a subterranean formation **110** located below the earth's surface **115**. The platform **120**, in at least one embodiment, has a hoisting apparatus **125** and a derrick **130** for raising and lowering a downhole conveyance **140**, such as a drill string, casing string, tubing string, coiled tubing, etc. Although a land-based oil and gas platform **120** is illustrated in FIG. 1, the scope of this disclosure is not thereby limited, and thus could potentially apply to offshore applications. The teachings of this disclosure may also be applied to other land-based multilateral wells different from that illustrated.

[0025] The well system **100** in one or more embodiments includes a main wellbore **150**. The main wellbore **150**, in the illustrated embodiment, includes tubing **160**, **165**, which may have differing tubular diameters. Extending from the main wellbore **150**, in one or more embodiments, may be one or more lateral wellbores **170**. Furthermore, a plurality of multilateral junctions **175** may be positioned at junctions between the main wellbore **150** and the lateral wellbores **170**. Each multilateral junction **175** may comprise a y-block and/or legs (e.g., mainbore leg and/or lateral bore leg) designed, manufactured or operated according to the disclosure. As discussed above, the multilateral junctions **175** may include the novel joints disclosed herein.

[0026] The well system **100** may additionally include the one or more ICVs **180** positioned at various positions within the main wellbore **150** and/or one or more of the lateral wellbores **170**. The ICVs **180** may comprise an ICV designed, manufactured or operated according to the disclosure. As discussed above, one or more of the ICVs **180** could include the novel joints disclosed herein, including in the sand control screen portion. The well system **100** may additionally include a control unit **190**. The control unit **190**, in this embodiment, is operable to provide control to or receive signals from, one or more downhole devices.

[0027] The well system may additionally include one or more packers **195** (e.g., zonal isolation production packers). The packers **195** may be designed, manufactured or operated according to the disclosure. As discussed above, one or more of the packers **195** could include the novel joints discussed herein.

[0028] Turning to FIG. 2, depicted is a cross-sectional view of a downhole tool **200** including a junction **205** designed, manufactured and operated according to one or more embodiments of the disclosure. The junction **205** of FIG. 2 includes a first member **210** and second member **220**. In accordance with one or more embodiments of the disclosure, the first member **210** comprises a first material (M1) and the second member **220** comprises a second material (M2). In certain embodiments, the first material (M1) and the second material (M2) are the same material, but in other embodiments the first material (M1) and the second material (M2) are different materials. In one or more embodiments, the first material (M1) and the second material (M2) comprise a crack compliant material. In at least one embodiment, the first material (M1) and the second material (M2) comprise a material having a yield strength of no greater than 140 ksi, if not no greater than 125 ksi, if not no greater than 110 ksi, if not no greater than 95 ksi, if not no greater than 80 ksi, if not no greater than 50 ksi.

[0029] In the illustrated embodiment, and in accordance with the disclosure, the first member **210** has a first member inside diameter (ID₁), a first member outside diameter (OD₁), a first member thickness (t₁), and a first member hardness value (HV₁). In the illustrated embodiment, and in accordance with the disclosure, the second member **220** has a second member inside diameter (ID₂), a second member outside diameter (OD₂), a second member thickness (t₂), and a second member hardness value (HV₂).

[0030] In accordance with one embodiment of the disclosure, the first member **210** and the second member **220** overlap one another. Depending on the design, the overlap may be face-to-face, end-to-end, butt-to-butt, or any other overlap, as well as combinations of the same. The first member **210** and the second member **220**, in the illustrated embodiment, thus define an overlapping space **230**. The overlapping space **230**, in at least one or more embodiments, defines the type of junction. For example, in the embodiment of FIG. 2, the overlapping space **230** is a radially extending space that would result in a butt joint. However, in other embodiments, the overlapping space **230** could be an angled radially extending space that would result in an angled butt joint. In even yet other embodiments, the overlapping space **230** could be a single stepped overlapping space (e.g., where a portion of one member extends within a portion of the other member), which would tend to form a single step overlapping joint, as further discussed below.

[0031] While not required, the first member **210** and the second member **220** may be a first tubular and a second tubular, such as in the embodiment discussed with regard to FIG. 2. Accordingly, the first member **210** and the second member **220** may define a centerline (C_L). In other embodiments, however, one or both of the first member **210** or the second member **220** are not tubulars. In at least one embodiment, the second member **220** is a collet (e.g., a pressure-containing collet) being coupled to the first member **210**.

[0032] In the illustrated embodiment, the first member **210** has a length (L₁) and the second member **220** has a length (L₂). In the illustrated embodiment, at least a portion of the overlapping space **230** (and thus the resulting metal joint) is substantially perpendicular with the length (L₁), thus forming a radially extending portion of the overlapping space **230**. As will be understood herein, the overlapping space may be substantially perpendicular to the centerline (C_L), or in certain other embodiments not perpendicular to the centerline (C_L) (e.g., substantially parallel with the centerline (C_L)). As will be discussed below, other embodiments exist wherein at least a portion of the overlapping space **230** (and thus the resulting metal joint) is substantially parallel with the length (L₁) (e.g., forming an axially extending portion of the overlapping space **230**) and/or angled relative to the length (L₁).

[0033] In the illustrated embodiment, the downhole tool **200** includes a crack compliant seal **240** positioned at the overlapping space **230**, the crack compliant seal **240** having a compliant seal hardness value (HV_{cs}) less than or equal to one or more of the first member hardness value (HV₁) or second member hardness value (HV₂). Accordingly, the crack compliant seal **240** is positioned in at least a portion of the radially extending portion of the overlapping space **230**. The crack compliant seal **240** may comprise any of the crack compliant materials discussed above and remain within the purview of the disclosure. In the illustrated embodiment, the crack compliant seal **240** is a crack com-

pliant brazed seal, and thus was positioned within the overlapping space **230** using a conventional brazing process. In other embodiments, the crack compliant seal **240** could be a crack compliant weld, or another type of crack compliant seal.

[0034] As indicated above, the compliant seal hardness value (HV_{cs}) may be a post weld heat treatment compliant seal hardness value (HV_{cs-pw}). Furthermore, in at least one embodiment, the post weld compliant seal hardness value (HV_{cs-pw}) may be **250** or less. In yet another embodiment, the post weld compliant seal hardness value (HV_{cs-pw}) may be **250** or less after being subjected to the post weld heat treatment of at least 620°C . In even another embodiment, the crack compliant seal may include no more than 1% mass fraction of nickel.

[0035] In the illustrated embodiment, a high yield strength weld **250** is also positioned at the overlapping space **230**. In accordance with the disclosure, the high yield strength weld **250** has a yield strength of at least 80 ksi. As discussed above, the high yield strength weld **250** could also have a yield strength of at least 95 ksi, a yield strength of at least 110 ksi, a yield strength of at least 125 ksi, a yield strength of at least 140 ksi, or even greater. The location of the high yield strength weld **250** relative to the crack compliant seal **240** may be dependent on the location of a source of corrosive fluid (e.g., H_2S gas, CO_2 gas, H_2 gas, etc.). For example, if the corrosive fluid were located on the inside of the first member **210** and the second member **220**, the crack compliant seal **240** would ideally be located radially inside of the high yield strength weld **250**, such as in the embodiment of FIG. 2.

[0036] Turning to FIG. 3, depicted is a cross-sectional view of a downhole tool **300** including a junction **305** designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **300** having the junction **305** is similar in many respects to the downhole tool **200** of FIG. 2. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **300** differs, for the most part, from the downhole tool **200**, in that the downhole tool **300** contemplates corrosive fluid additionally on the outside of the first member **210** and the second member **220**. Accordingly, in the embodiment of FIG. 3, a second crack compliant seal **340** is located radially outside of the high yield strength weld **250**. In one or more embodiments, the second crack compliant seal **340** is a crack compliant weld, as shown.

[0037] Turning to FIG. 4, depicted is a cross-sectional view of a downhole tool **400** including a junction **405** designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **400** having the junction **405** is similar in many respects to the downhole tool **200** of FIG. 2. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **400** differs, for the most part, from the downhole tool **200**, in that the downhole tool **400** contemplates corrosive fluid only on the outside of the first member **210** and the second member **220**. Accordingly, in the embodiment of FIG. 4, a crack compliant seal **440** is located radially outside of the high yield strength weld **450**. In one or more embodiments, the crack compliant seal **440** is a crack compliant weld, as shown.

[0038] Turning to FIG. 5, depicted is a cross-sectional view of a downhole tool **500** including a junction **505**

designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **500** having the junction **505** is similar in many respects to the downhole tool **200** of FIG. 2. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **500** differs, for the most part, from the downhole tool **200**, in that the second member outside diameter (OD_2) is less than the first member inside diameter (ID_1), the second member **520** having slid into the first member **510** to form an overlapping space (e.g., ultimately resulting in an overlapping joint). Further to the embodiment of FIG. 5, the downhole tool **500** employs a single stepped overlapping space **530**. The single stepped overlapping space **530**, in the illustrated embodiment, includes a radial interior radial extending portion **530a**, an axial extending portion **530b**, and a radial exterior radial extending portion **530c**.

[0039] In the embodiment of FIG. 5, the crack compliant seal **540** is at least partially located within the axial extending portion **530b** of the overlapping space **530**, and the high yield strength weld **550** is at least partially located within the radial exterior radial extending portion **530c** of the overlapping space **530**. While not shown, the crack compliant seal **540** could also be located within at least a portion of the radial interior radial extending portion **530a**, or alternatively or in addition to, within at least a portion of the radial exterior radial extending portion **530c**.

[0040] Turning to FIG. 6, depicted is a cross-sectional view of a downhole tool **600** including a junction **605** designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **600** having the junction **605** is similar in many respects to the downhole tool **500** of FIG. 5. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **600** differs, for the most part, from the downhole tool **500**, in that the downhole tool **600** contemplates corrosive fluid both inside the first member **510** and the second member **520** and outside of the first member **510** and the second member **520**. Accordingly, a sandwich of a first crack compliant seal **640a**, a high yield strength weld **650**, and a second crack compliant seal **640b** (e.g., crack compliant weld) is employed.

[0041] Turning to FIG. 7, depicted is a cross-sectional view of a downhole tool **700** including a junction **705** designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **700** having the junction **705** is similar in many respects to the downhole tool **500** of FIG. 5. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **700** differs, for the most part, from the downhole tool **500**, in that the downhole tool **700** contemplates corrosive fluid only on the outside of the first member **510** and the second member **520**. Accordingly, in the embodiment of FIG. 7, a crack compliant seal **740** is located radially outside of the high yield strength weld **750**. In one or more embodiments, the crack compliant seal **740** is a crack compliant weld, as shown.

[0042] Turning to FIGS. 8A through 8C, depicted are cross-sectional views of a downhole tool **800** including a junction **805** designed, manufactured and operated according to one or more alternative embodiments of the disclosure at different stages of manufacture. The downhole tool **800** having the junction **805** is similar in many respects to the downhole tool **500** of FIG. 5. Accordingly, like reference

numbers have been used to indicate similar, if not identical, features. The downhole tool **800** differs, for the most part, from the downhole tool **500**, in that the downhole tool **800** includes a radially exterior notched overlapping junction. For example, each of the first member **510** and the second member **530** could have angled surfaces forming the notch **810**. It should be noted that the notch **810** may exist as a function of the original shapes of the first member **510** and the second member **520**, or may be formed (e.g., cut, ground, etc.) from the first member **510** and the second member **520** after their initial joinder.

[0043] Further to the embodiment of FIGS. **8A** through **8C**, a crack compliant weld **840** may be first formed (e.g., deposited) in the notch **810**, followed by the forming of the high yield string weld **850** (e.g., at least partially within the notch **810**), as shown in FIGS. **8B** and **8C**. In the illustrated embodiment, the crack compliant weld **840** is located radially inside of the high yield strength weld **850**, as might be desired/required if the corrosive fluid were located inside of the first member **510** and the second member **520**.

[0044] Turning to FIG. **9**, depicted is a cross-sectional view of a downhole tool **900** including a junction **905** designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **900** having the junction **905** is similar in many respects to the downhole tool **800** of FIG. **8**. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **900** differs, for the most part, from the downhole tool **800**, in that the downhole tool **900** contemplates corrosive fluid both inside the first member **510** and the second member **520** and outside of the first member **510** and the second member **520**. Accordingly, a sandwich of a first crack compliant seal **940a**, a high yield strength weld **950**, and a second crack compliant weld **940b** is employed.

[0045] Turning to FIG. **10**, depicted is a cross-sectional view of a downhole tool **1000** including a junction **1005** designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **1000** having the junction **1005** is similar in many respects to the downhole tool **800** of FIG. **8**. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **1000** differs, for the most part, from the downhole tool **800**, in that the downhole tool **1000** contemplates corrosive fluid only on the outside of the first member **510** and the second member **520**. Accordingly, in the embodiment of FIG. **10**, a crack compliant seal **1040** is located radially outside of the high yield strength weld **1050**. In one or more embodiments, the crack compliant seal **1040** is a crack compliant weld **1050**, as shown.

[0046] Turning to FIG. **11**, depicted is a cross-sectional view of a downhole tool **1100** including a junction **1105** designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **1100** having the junction **1105** is similar in many respects to the downhole tool **800** of FIG. **8**. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **1100** differs, for the most part, from the downhole tool **800**, in that the downhole tool **1100** includes a geometric mechanical strengthening feature **1170** located between the first member **510** and the second member **520**. The geometric mechanical strengthening feature **1170**, in the disclosed embodiment, is

configured to increase an engineering rating (e.g., as defined above) of the overlapping space **530**, and thus the joint. For instance, in certain embodiments, the geometric mechanical strengthening feature **1170** may provide the necessary yield strength for the downhole tool **1100**, such that the high yield strength weld may not be necessary.

[0047] Many different geometric mechanical strengthening features may be used and remain within the scope of the disclosure. Nevertheless, in the embodiment of FIG. **11**, the geometric mechanical strengthening feature **1170** is a first set of threads **1170a** in the first member inside diameter (ID_1) and a second set of threads **1170b** in the second member outside diameter (OD_2). In this embodiment, the collection of the first set of threads **1170a** and the second set of threads **1170b** provides the requisite yield strength for the joint. Further to the embodiment of FIG. **11**, a crack compliant seal **1140** is positioned at the overlapping space **530**. Further to the embodiment of FIG. **11**, the crack compliant seal **1140** may be a crack compliant weld.

[0048] Turning to FIG. **12**, depicted is a cross-sectional view of a downhole tool **1200** including a junction **1205** designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **1200** having the junction **1205** is similar in many respects to the downhole tool **1100** of FIG. **11**. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **1200** differs, for the most part, from the downhole tool **1100**, in that the downhole tool **1200** contemplates corrosive fluid both inside the first member **510** and the second member **520** and outside of the first member **510** and the second member **520**. Accordingly, a sandwich of a first crack compliant seal **1240a**, a high yield strength weld **1250**, and a second crack compliant weld **1240b** is employed. The first crack compliant seal **1240** may be located within an axial extending portion of the overlapping joint, located between the first set of threads and the second set of threads, or located within a radial extending portion of the overlapping joint, depending on the design of the downhole tool **1200**.

[0049] Turning to FIG. **13**, depicted is a cross-sectional view of a downhole tool **1300** including a junction **1305** designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **1300** having the junction **1305** is similar in many respects to the downhole tool **1100** of FIG. **11**. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **1300** differs, for the most part, from the downhole tool **1100**, in that the downhole tool **1300** further includes a high yield strength weld **1350** positioned at the overlapping space **530**. In the illustrated embodiment, the crack compliant seal **1140** is located radially inside of the high yield strength weld **1350**, the high yield strength weld **1350** providing additional burst and/or collapse support. The location of the crack compliant seal **1140** is, at least in part, based upon the corrosive fluid being location on an inside of the of the first member **510** and the second member **520**.

[0050] Turning to FIG. **14**, depicted is a cross-sectional view of a downhole tool **1400** including a junction **1405** designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **1400** having the junction **1405** is similar in many respects to the downhole tool **1100** of FIG. **11**. Accordingly, like reference numbers have been used to indicate similar, if

not identical, features. The downhole tool **1400** differs, for the most part, from the downhole tool **1100**, in that the downhole tool **1400** further includes a high yield strength weld **1450** positioned at the overlapping space **530**. In the illustrated embodiment, the crack compliant seal **1440** is located radially outside of the high yield strength weld **1450**, the high yield strength weld **1450** providing additional burst and/or collapse support. The location of the crack compliant seal **1440** is, at least in part, based upon the corrosive fluid being location on an outside of the of the first member **510** and the second member **520**.

[0051] Turning to FIG. **15**, depicted is a cross-sectional view of a downhole tool **1500** including a junction **1505** designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **1500** having the junction **1505** is similar in many respects to the downhole tool **800** of FIG. **8**. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **1500** differs, for the most part, from the downhole tool **800**, in that the downhole tool **1500** includes a geometric mechanical strengthening feature **1570** located between the first member **510** and the second member **520**. The geometric mechanical strengthening feature **1570**, in the disclosed embodiment, is configured to increase an engineering rating (e.g., as defined above) of the overlapping space **530**. For instance, in certain embodiments, the geometric mechanical strengthening feature **1570** may provide the necessary yield strength for the downhole tool such that the high yield strength weld may not be necessary.

[0052] Many different geometric mechanical strengthening features may be used and remain within the scope of the disclosure, including collets, lock rings, snap rings, etc. Nevertheless, in the embodiment of FIG. **15**, the geometric mechanical strengthening feature **1570** is a collet feature in one of the first member **510** or the second member **520**. Further to the embodiment of FIG. **15**, the collet feature may include a collet finger profile **1580a** in one of the first member **510** or the second member **520** and a collet finger **1580b** in an other of the second member **520** or the first member **510**. In the illustrated embodiment of FIG. **15**, the collet finger profile **1580b** is located in the first member **510**, whereas the collet finger **1580b** is located in the second member **520**. Nevertheless, the opposite could hold true.

[0053] Further to the embodiment of FIG. **15**, a crack compliant seal **1540** is positioned at the overlapping space **530**. Further to the embodiment of FIG. **15**, the crack compliant seal **1540** may be a crack compliant weld.

[0054] Turning to FIG. **16**, depicted is a cross-sectional view of a downhole tool **1600** including a junction **1605** designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **1600** having the junction **1605** is similar in many respects to the downhole tool **1500** of FIG. **15**. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **1600** differs, for the most part, from the downhole tool **1500**, in that the downhole tool **1600** contemplates corrosive fluid both inside the first member **510** and the second member **520** and outside of the first member **510** and the second member **520**. Accordingly, a sandwich of a first crack compliant seal **1640a**, a high yield strength weld **1650**, and a second crack compliant weld **1640b** is employed.

[0055] Turning to FIG. **17**, depicted is a cross-sectional view of a downhole tool **1700** including a junction **1705** designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **1700** having the junction **1705** is similar in many respects to the downhole tool **1500** of FIG. **15**. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **1700** differs, for the most part, from the downhole tool **1500**, in that the downhole tool **1700** further includes a high yield strength weld **1750** positioned at the overlapping space **530**. In the illustrated embodiment, the crack compliant seal **1740** is located radially inside of the high yield strength weld **1750**, the high yield strength weld **1750** providing additional burst and/or collapse support. The location of the crack compliant seal **1740** is, at least in part, based upon the corrosive fluid being location on an inside of the of the first member **510** and the second member **520**.

[0056] Turning to FIG. **18**, depicted is a cross-sectional view of a downhole tool **1800** including a junction **1805** designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **1800** having the junction **1805** is similar in many respects to the downhole tool **1500** of FIG. **15**. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **1800** differs, for the most part, from the downhole tool **1500**, in that the downhole tool **1800** further includes a high yield strength weld **1850** positioned at the overlapping space **530**. In the illustrated embodiment, the crack compliant seal **1840** is located radially outside of the high yield strength weld **1850**, the high yield strength weld **1850** providing additional burst and/or collapse support. The location of the crack compliant seal **1840** is, at least in part, based upon the corrosive fluid being location on an outside of the of the first member **510** and the second member **520**.

[0057] Turning to FIG. **19**, depicted is a cross-sectional view of a downhole tool **1900** including a junction **1905** designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **1900** having the junction **1905** is similar in many respects to the downhole tool **1500** of FIG. **15**. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **1900** differs, for the most part, from the downhole tool **1500**, in that the downhole tool **1900** further includes a geometric mechanical strengthening feature **1970** including a second set of collet finger profiles **1980a** and collet fingers **1980b**. In one or more embodiments, the second set of collet finger profiles **1980a** and collet fingers **1980b** provide additional burst/collapse strength. a high yield strength weld **1850** positioned at the overlapping space **530**.

[0058] Turning to FIG. **20**, depicted is a cross-sectional view of a downhole tool **2000** including a junction **2005** designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **2000** having the junction **2005** is similar in many respects to the downhole tool **1900** of FIG. **19**. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **2000** differs, for the most part, from the downhole tool **1900**, in that the downhole tool **2000** contemplates corrosive fluid both inside the first member **510** and the second member **520** and outside of the first member **510** and the second member **520**.

Accordingly, a sandwich of a first crack compliant seal **2040a**, a high yield strength weld **2050**, and a second crack compliant weld **2040b** is employed. In at least one embodiment, the first crack compliant seal **2040a** is located axially between the collet finger **1580b** and collet finger **1980b**.

[0059] Turning to FIG. **21**, depicted is a cross-sectional view of a downhole tool **2100** including a junction **2105** designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **2100** having the junction **2105** is similar in many respects to the downhole tool **1900** of FIG. **19**. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **2100** differs, for the most part, from the downhole tool **1900**, in that the downhole tool **2100** further includes a high yield strength weld **2150** positioned at the overlapping space **530**. In the illustrated embodiment, the crack compliant seal **2140** is located radially inside of the high yield strength weld **2150**, the high yield strength weld **2150** providing additional burst and/or collapse support. The location of the crack compliant seal **2140** is, at least in part, based upon the corrosive fluid being location on an inside of the of the first member **510** and the second member **520**.

[0060] Turning to FIG. **22**, depicted is a cross-sectional view of a downhole tool **2200** including a junction **2205** designed, manufactured and operated according to one or more alternative embodiments of the disclosure. The downhole tool **2200** having the junction **2205** is similar in many respects to the downhole tool **1900** of FIG. **19**. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The downhole tool **2200** differs, for the most part, from the downhole tool **1900**, in that the downhole tool **2200** further includes a high yield strength weld **2250** positioned at the overlapping space **530**. In the illustrated embodiment, the crack compliant seal **2240** is located radially outside of the high yield strength weld **2250**, the high yield strength weld **2250** providing additional burst and/or collapse support. The location of the crack compliant seal **2240** is, at least in part, based upon the corrosive fluid being location on an outside of the of the first member **510** and the second member **520**.

[0061] Turning to FIG. **23**, illustrated is a multilateral junction **2300** designed, manufactured and operated according to one or more embodiments of the disclosure. The multilateral junction **2300** includes a y-block **2310**. In accordance with one or more embodiments of the disclosure, the y-block **2310** includes a housing **2320** having a first end **2322** and a second opposing end **2324**. The housing **2320**, without limitation, may comprise steel or another suitable material.

[0062] Extending into the housing **2320** from the first end **2322** is a single first bore **2330**. The single first bore **2330**, in accordance with one embodiment, defines a first centerline **2335**. The y-block **2310** additionally includes second and third separate bores **2340**, **2350**, respectively, extending into the housing **2320** and branching off from the single first bore **2330**. In accordance with one or more embodiments, the second bore **2340** defines a second centerline **2345**, and the third bore **2350** defining a third centerline **2355**.

[0063] The multilateral junction **2300**, as illustrated in FIG. **23**, additionally includes a mainbore leg **2360** coupled to the second bore **2340** for extending into the main wellbore. In at least one embodiment, the mainbore leg **2360** and the second bore **2340** define a second overlapping space

2365. The multilateral junction **2300**, as illustrated in FIG. **23**, additionally includes a lateral bore leg **2370** coupled to the third bore **2350** for extending into the lateral wellbore. In at least one embodiment, the lateral bore leg **2370** and the third bore **2350** define a third overlapping space **2375**. In at least one embodiment, one or both of the lateral bore leg **2370** or the main bore leg **2360** is an approximately D-shaped tube.

[0064] The multilateral junction **2300**, in one or more embodiments, additionally includes a joint **2380** located in at least a portion of the second overlapping space **2365** or the third overlapping space **2375**, the joint **2380** designed, manufactured and/or employed according to any one of the embodiments discussed above. In at least one embodiment, the joint **2380** is a lateral wellbore leg joint **2382** located in at least a portion of the third overlapping space **2375**. In yet another embodiment, the joint **2380** is a main wellbore leg joint **2384** located in at least a portion of the second overlapping space **2365**. In yet another embodiment, both the lateral wellbore leg joint **2382** and the main wellbore leg joint **2384** exist.

[0065] The multilateral junction **2300**, in one or more embodiments, may additionally include a joint **2386** located in at least a portion of the single first bore **2330**. For example, the joint **2386** may be used to couple an additional tubular to the single first bore **2330**. In accordance with the disclosure, the joint **2386** may be designed, manufactured and/or employed according to any one of the embodiments discussed above.

[0066] It should also be noted that in certain other embodiments, the joints **2380** may be located in other portions of the multilateral junction **2300**. For instance, a seal stinger could be coupled at the end of the mainbore leg **2360**. In this embodiment, the joint **2380** may be used to couple the mainbore leg **2360** and the seal stinger. In another embodiment, a transition cross-over (e.g., D to round transition cross-over) could be coupled at the end of the lateral bore leg **2370**. In this embodiment, the joint **2380** may be used to couple the lateral bore leg **2370** to the transition cross-over.

[0067] It should further be noted that even though FIG. **23** is directed to a downhole tool comprising a multilateral junction, in other embodiments the downhole tool could comprise a downhole packer, among other features.

[0068] Aspects disclosed herein include:

[0069] A. A downhole tool, the downhole tool including: 1) a first member, the first member having a first member inside diameter (ID_1), a first member outside diameter (OD_1), a first member thickness (t_1), and a first member hardness value (HV_1); 2) a second member, the second member having a second member inside diameter (ID_2), a second member outside diameter (OD_2), a second member thickness (t_2), and a second member hardness value (HV_2); the first and second members positioned proximate one another forming an overlapping space therebetween; 3) a crack compliant seal positioned at the overlapping space, the crack compliant seal having a compliant seal hardness value (HV_{cs}) less than or equal to one or more of the first member hardness value (HV_1) and second member hardness value (HV_2); and 4) a high yield strength weld positioned at the overlapping space, the high yield strength weld having a yield strength of at least 50 ksi.

[0070] B. A well system, the well system including: 1) a wellbore; 2) a downhole tool positioned within the

wellbore, the downhole tool including: a) a first member, the first member having a first member inside diameter (ID_1), a first member outside diameter (OD_1), a first member thickness (t_1), and a first member hardness value (HV_1); b) a second member, the second member having a second member inside diameter (ID_2), a second member outside diameter (OD_2), a second member thickness (t_2), and a second member hardness value (HV_2) the first and second members positioned proximate one another forming an overlapping space therebetween; c) a crack compliant seal positioned at the overlapping space, the crack compliant seal having a compliant seal hardness value (HV_{cs}) less than or equal to one or more of the first member hardness value (HV_1) or second member hardness value (HV_2); and d) a high yield strength weld positioned at the overlapping space, the high yield strength weld having a yield strength of at least 50 ksi.

[0071] C. A method, the method including: 1) obtaining a first member, the first member having a first member inside diameter (ID_1), a first member outside diameter (OD_1), a first member thickness (t_1), and a first member hardness value (HV_1); 2) obtaining a second member, the second member having a second member inside diameter (ID_2), a second member outside diameter (OD_2), a second member thickness (t_2), and a second member hardness value (HV_2); 3) positioning the first and second members proximate one another forming an overlapping space therebetween, wherein a crack compliant seal having a compliant seal hardness value (HV_{cs}) less than or equal to one or more of the first member hardness value (HV_1) or second member hardness value (HV_2) is positioned at the overlapping space, and a high yield strength weld is positioned at the overlapping space, the high yield strength weld having a yield strength of at least 50 ksi.

[0072] D. A downhole tool, the downhole tool including: 1) a first member, the first member having a first member inside diameter (ID_1), a first member outside diameter (OD_1), a first member thickness (t_1), and a first member hardness value (HV_1); 2) a second member, the second member having a second member inside diameter (ID_2), a second member outside diameter (OD_2), a second member thickness (t_2), and a second member hardness value (HV_2), the first and second members positioned proximate one another forming an overlapping space therebetween; 3) a geometric mechanical strengthening feature located between the first member and the second member, the geometric mechanical strengthening feature configured to increase an engineering rating of the overlapping space; and 4) a crack compliant seal positioned at the overlapping space, the crack compliant seal having a compliant seal hardness value (HV_{cs}) less than or equal to one or more of the first member hardness value (HV_1) or second member hardness value (HV_2).

[0073] E. A well system, the well system including: 1) a wellbore; 2) a downhole tool positioned within the wellbore, the downhole tool including: a) a first member, the first member having a first member inside diameter (ID_1), a first member outside diameter (OD_1), a first member thickness (t_1), and a first member hardness value (HV_1); b) a second member, the second member having a second member inside diameter

(ID_2), a second member outside diameter (OD_2), a second member thickness (t_2), and a second member hardness value (HV_2) the first and second members positioned proximate one another forming an overlapping space therebetween; c) a geometric mechanical strengthening feature located between the first member and the second member, the geometric mechanical strengthening feature configured to increase an engineering rating of the overlapping space; and d) a crack compliant seal positioned at the overlapping space, the crack compliant seal having a compliant seal hardness value (HV_{cs}) less than or equal to one or more of the first member hardness value (HV_1) or second member hardness value (HV_2).

[0074] F. A method, the method including: 1) obtaining a first member, the first member having a first member inside diameter (ID_1), a first member outside diameter (OD_1), a first member thickness (t_1), and a first member hardness value (HV_1); 2) obtaining a second member, the second member having a second member inside diameter (ID_2), a second member outside diameter (OD_2), a second member thickness (t_2), and a second member hardness value (HV_2); 3) coupling the first and second members together using a geometric mechanical strengthening feature forming an overlapping space therebetween, wherein a crack compliant seal is positioned at the overlapping space, the crack compliant seal having a compliant seal hardness value (HV_{cs}) less than or equal to one or more of the first member hardness value (HV_1) or second member hardness value (HV_2).

[0075] Aspects A, B, C, D, E, and F may have one or more of the following additional elements in combination: Element 1: wherein the compliant seal hardness value (HV_{cs}) is a post weld heat treatment compliant seal hardness value (HV_{cs-pw}). Element 2: wherein the post weld compliant seal hardness value (HV_{cs-pw}) is 250 or less. Element 3: wherein the post weld compliant seal hardness value (HV_{cs-pw}) is 250 or less after being subjected to the post weld heat treatment of at least 620° C. Element 4: wherein the crack compliant seal includes no more than 1% mass fraction of nickel. Element 5: wherein the crack compliant seal is located radially inside of the high yield strength weld. Element 6: wherein the crack compliant seal is a first crack compliant seal, and further including a second crack compliant seal located radially outside of the high yield strength weld. Element 7: wherein the second crack compliant seal is a second crack compliant weld. Element 8: wherein the crack compliant seal is located radially outside of the high yield strength weld. Element 9: wherein the crack compliant seal is a crack compliant weld. Element 10: wherein the overlapping space is a butt joint. Element 11: wherein the second member outside diameter (OD_2) is less than the first member inside diameter (ID_1), the second member having slid into the first member to form an overlapping joint. Element 12: wherein the crack compliant seal is at least partially located within an axial extending portion of the overlapping joint and the high yield strength weld is at least partially located within a radial extending portion of the overlapping joint. Element 13: wherein the crack compliant seal is a first crack compliant seal, and further including a second crack compliant seal located radially outside of the high yield strength weld. Element 14: further including a geometric mechanical strengthening feature located between the

first member and the second member. Element 15: wherein the geometric mechanical strengthening feature is a first set of threads in the first member inside diameter (ID_1) and a second set of threads in the second member outside diameter (OD_2). Element 16: wherein the geometric mechanical strengthening feature is a collet feature in one of the first member or the second member. Element 17: wherein the first member forms at least a portion of a y-block of a multilateral junction and the second member forms at least a portion of a multilateral mainbore leg or lateral bore leg. Element 18: wherein the first member and the second member form at least a portion of downhole packer. Element 19: wherein the high yield strength weld has a yield strength of at least 80 ksi. Element 20: wherein the first member and second member form at least a portion of a downhole tool, and further including positioning the downhole tool including the crack compliant seal and the high yield strength weld within a wellbore. Element 21: wherein the geometric mechanical strengthening feature is a collet feature in one of the first member or the second member forming an overlapping joint. Element 22: wherein the collet feature includes a collet finger in one of the first member or the second member and a collet finger profile in an other of the second member or the first member, and further wherein the crack compliant material is axially located between the collet finger profile and a radial extending portion of the overlapping joint. Element 23: wherein the geometric mechanical strengthening feature is a first set of threads in the first member inside diameter (ID_1) and a second set of threads in the second member outside diameter (OD_2) forming an overlapping joint. Element 24: wherein the crack compliant material is located within an axial extending portion of the overlapping joint. Element 25: wherein the crack compliant material is located between the first set of threads and the second set of threads. Element 26: wherein the crack compliant material is located within a radial extending portion of the overlapping joint. Element 27: further including a high yield strength weld positioned at the overlapping space radially outside of the crack compliant material, the high yield strength weld having a yield strength of at least 50 ksi. Element 28: wherein the crack compliant seal is a first crack compliant seal, and further including a second crack compliant seal located radially outside of the high yield strength weld. Element 29: wherein the high yield strength weld is located within a radial extending portion of the overlapping joint.

[0076] Those skilled in the art to which this application relates will appreciate that other and further additions, deletions, substitutions and modifications may be made to the described embodiments.

What is claimed is:

1. A downhole tool, comprising:

- a first member, the first member having a first member inside diameter (ID_1), a first member outside diameter (OD_1), a first member thickness (t_1), and a first member hardness value (HV_1);
- a second member, the second member having a second member inside diameter (ID_2), a second member outside diameter (OD_2), a second member thickness (t_2), and a second member hardness value (HV_2), the first and second members positioned proximate one another forming an overlapping space therebetween;
- a geometric mechanical strengthening feature located between the first member and the second member, the

geometric mechanical strengthening feature configured to increase an engineering rating of the overlapping space; and

- a crack compliant seal positioned at the overlapping space, the crack compliant seal having a compliant seal hardness value (HV_{cs}) less than or equal to one or more of the first member hardness value (HV_1) or second member hardness value (HV_2).

2. The downhole tool as recited in claim 1, wherein the compliant seal hardness value (HV_{cs}) is a post weld heat treatment compliant seal hardness value (HV_{cs-pw}).

3. The downhole tool as recited in claim 2, wherein the post weld compliant seal hardness value (HV_{cs-pw}) is 250 or less.

4. The downhole tool as recited in claim 3, wherein the post weld compliant seal hardness value (HV_{cs-pw}) is 250 or less after being subjected to the post weld heat treatment of at least 620° C.

5. The downhole tool as recited in claim 4, wherein the crack compliant seal includes no more than 1% mass fraction of nickel.

6. The downhole tool as recited in claim 1, wherein the geometric mechanical strengthening feature is a collet feature in one of the first member or the second member forming an overlapping joint.

7. The downhole tool as recited in claim 6, wherein the collet feature includes a collet finger in one of the first member or the second member and a collet finger profile in an other of the second member or the first member, and further wherein the crack compliant material is axially located between the collet finger profile and a radial extending portion of the overlapping joint.

8. The downhole tool as recited in claim 1, wherein the geometric mechanical strengthening feature is a first set of threads in the first member inside diameter (ID_1) and a second set of threads in the second member outside diameter (OD_2) forming an overlapping joint.

9. The downhole tool as recited in claim 8, wherein the crack compliant material is located within an axial extending portion of the overlapping joint.

10. The downhole tool as recited in claim 8, wherein the crack compliant material is located between the first set of threads and the second set of threads.

11. The downhole tool as recited in claim 8, wherein the crack compliant material is located within a radial extending portion of the overlapping joint.

12. The downhole tool as recited in claim 1, further including a high yield strength weld positioned at the overlapping space radially outside of the crack compliant material, the high yield strength weld having a yield strength of at least 50 ksi.

13. The downhole tool as recited in claim 12, wherein the crack compliant seal is a first crack compliant seal, and further including a second crack compliant seal located radially outside of the high yield strength weld.

14. The downhole tool as recited in claim 13, wherein the high yield strength weld is located within a radial extending portion of the overlapping joint.

15. The downhole tool as recited in claim 12, wherein the high yield strength weld has a yield strength of at least 80 ksi.

16. A well system, comprising:

a wellbore;

a downhole tool positioned within the wellbore, the downhole tool including:

a first member, the first member having a first member inside diameter (ID_1), a first member outside diameter (OD_1), a first member thickness (t_1), and a first member hardness value (HV_1);

a second member, the second member having a second member inside diameter (ID_2), a second member outside diameter (OD_2), a second member thickness (t_2), and a second member hardness value (HV_2) the first and second members positioned proximate one another forming an overlapping space therebetween;

a geometric mechanical strengthening feature located between the first member and the second member, the geometric mechanical strengthening feature configured to increase an engineering rating of the overlapping space; and

a crack compliant seal positioned at the overlapping space, the crack compliant seal having a compliant seal hardness value (HV_{cs}) less than or equal to one or more of the first member hardness value (HV_1) or second member hardness value (HV_2).

17. The well system as recited in claim 16, wherein the compliant seal hardness value (HV_{cs}) is a post weld heat treatment compliant seal hardness value (HV_{cs-pw}).

18. The well system as recited in claim 17, wherein the post weld compliant seal hardness value (HV_{cs-pw}) is 250 or less.

19. The well system as recited in claim 18, wherein the post weld compliant seal hardness value (HV_{cs-pw}) is 250 or less after being subjected to the post weld heat treatment of at least 620° C.

20. The well system tool as recited in claim 19, wherein the crack compliant seal includes no more than 1% mass fraction of nickel.

21. The well system as recited in claim 16, wherein the geometric mechanical strengthening feature is a collet feature in one of the first member or the second member forming an overlapping joint.

22. The well system as recited in claim 21, wherein the collet feature includes a collet finger in one of the first member or the second member and a collet finger profile in an other of the second member or the first member, and further wherein the crack compliant material is axially located between the collet finger profile and a radial extending portion of the overlapping joint.

23. The well system as recited in claim 16, wherein the geometric mechanical strengthening feature is a first set of threads in the first member inside diameter (ID_1) and a second set of threads in the second member outside diameter (OD_2) forming an overlapping joint.

24. The well system as recited in claim 23, wherein the crack compliant material is located within an axial extending portion of the overlapping joint.

25. The well system as recited in claim 23, wherein the crack compliant material is located between the first set of threads and the second set of threads.

26. The well system as recited in claim 23, wherein the crack compliant material is located within a radial extending portion of the overlapping joint.

27. The well system as recited in claim 16, further including a high yield strength weld positioned at the overlapping space radially outside of the crack compliant material, the high yield strength weld having a yield strength of at least 50 ksi.

28. The well system as recited in claim 27, wherein the crack compliant seal is a first crack compliant seal, and further including a second crack compliant seal located radially outside of the high yield strength weld.

29. The well system as recited in claim 28, wherein the high yield strength weld is located within a radial extending portion of the overlapping joint.

30. The well system as recited in claim 27, wherein the high yield strength weld has a yield strength of at least 80 ksi.

31. A method, comprising:

obtaining a first member, the first member having a first member inside diameter (ID_1), a first member outside diameter (OD_1), a first member thickness (t_1), and a first member hardness value (HV_1);

obtaining a second member, the second member having a second member inside diameter (ID_2), a second member outside diameter (OD_2), a second member thickness (t_2), and a second member hardness value (HV_2); and

coupling the first and second members together using a geometric mechanical strengthening feature forming an overlapping space therebetween, wherein a crack compliant seal is positioned at the overlapping space, the crack compliant seal having a compliant seal hardness value (HV_{cs}) less than or equal to one or more of the first member hardness value (HV_1) or second member hardness value (HV_2).

32. The method as recited in claim 31, wherein the compliant seal hardness value (HV_{cs}) is a post weld heat treatment compliant seal hardness value (HV_{cs-pw}).

33. The method as recited in claim 32, wherein the post weld compliant seal hardness value (HV_{cs-pw}) is 250 or less.

34. The method as recited in claim 33, wherein the post weld compliant seal hardness value (HV_{cs-pw}) is 250 or less after being subjected to the post weld heat treatment of at least 620° C.

35. The method tool as recited in claim 34, wherein the crack compliant seal includes no more than 1% mass fraction of nickel.

36. The method as recited in claim 31, wherein the first member and second member form at least a portion of a downhole tool, and further including positioning the downhole tool including the crack compliant seal within a wellbore.

37. The method as recited in claim 31, wherein the first member forms at least a portion of a y-block of a multilateral junction and the second member forms at least a portion of a multilateral mainbore leg or lateral bore leg.

38. The method as recited in claim 31, wherein the first member and the second member form at least a portion of downhole packer.

39. The method as recited in claim 31, wherein the mechanical strengthening feature is a collet feature in one of the first member or the second member forming an overlapping joint.

40. The method as recited in claim 31, wherein the mechanical strengthening feature is a first set of threads in the first member inside diameter (ID₁) and a second set of threads in the second member outside diameter (OD₂) forming an overlapping joint.

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